



A survey of large language models: techniques, applications, and challenges

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Abstract

Amid the rapid advancement of artificial intelligence (AI) technology, large language models (LLMs) have emerged as a pivotal force driving innovation in the field, attracting researchers due to their distinct advantages and broad application potential. Unlike existing surveys that focus on technical enumeration or single-dimensional evaluation, this work develops an integrated analytical framework of “three-dimensional trade-off + scenario adaptation”. Centered on the dynamic balance of “performance-efficiency-cost”, this framework links model architectural selection, training strategy optimization, and deployment implementation, forming a closed-loop analytical logic that bridges technical characteristics and scenario requirements. This survey presents a systematic and comprehensive overview of LLMs, articulating their evolution through a lens of performance-efficiency-cost trade-offs. Unlike prior surveys, we place a particular emphasis on the comparative analysis of architectural paradigms (Encoder-Only, Decoder-Only, Encoder-Decoder) and training strategies (e.g., RLHF vs. SFT), providing a practical framework for selecting appropriate technologies for specific application needs. We further synthesize the landscape by delving into emerging paradigms such as Retrieval-Augmented Generation (RAG) and AI Agents, and critically examine pressing challenges including hallucination, scalability, and ethical dilemmas. By integrating foundational technical reviews with forward-looking trend and governance analysis, this work establishes a decision-oriented reference framework—guiding researchers in prioritizing research directions and assisting practitioners in matching technical solutions to specific scenarios—filling the gap of ‘overemphasizing technical description while lacking practical decision support’ in prior surveys. The unique value of this work, distinguishing it from prior literature, resides in three interconnected aspects: First, it takes “performance-efficiency-cost” trade-offs as the core clue to link architectural selection, training strategies, and deployment solutions (see Tables 2, 4), developing actionable technical selection logic for scenario-specific resource constraints. Second, it integrates the synergy between emerging paradigms (RAG, AI Agent) and traditional LLMs—for example, explaining how RAG reduces Decoder-Only models hallucination rate from 15.8% to 8.3% (Sects. 2.5.6, 4.2) and how AI Agent extends LLMs’ tool-use capabilities via transfer learning (Sect. 3.6) [1]. As a pure LLM, BioBERT outperforms untuned multimodal LLMs in text-aided low-resolution medical image recognition (e.g., matching image-based lesion features with text-derived diagnostic keywords).

Keywords Natural language processing · Computer vision · Cross-modal tasks · Data scarcity and annotation costs

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1 Introduction

LLMs mark a paradigm shift in artificial intelligence (AI), enabling robust generalization across diverse domains without task-specific architectures. Driven by advances in computing power and the explosive growth of data resources, LLMs have achieved remarkable progress in traditional domains such as natural language processing, computer vision, and speech recognition [2, 3].

Despite these advancements, LLM development is confronted with significant challenges. As model scales expand,

issues such as excessive computational resource consumption, low training efficiency, and poor interpretability have become critical bottlenecks [4]. Cross-modal task applications face hurdles in information fusion and generalization enhancement [5]. More critically, ethical dilemmas such as privacy risks [6], data security concerns, and algorithmic discrimination have become increasingly prominent, demanding urgent attention [7].

This study provides a comprehensive and in-depth analysis of LLMs, starting with foundational theories and frameworks. It systematically reviews key technological advancements, including core components like attention mechanisms [2], pre-training fine-tuning paradigms [3], and Transformer architectures. The paper delves into application status across various fields, with a particular focus on challenges and opportunities in cross-modal tasks [5]. The study also examines societal impacts and proposes mitigation strategies, building on foundational advances in few-shot learning (FSL) [8].

Existing LLM surveys primarily focus on single-dimensional technical enumeration, but this work differs in three critical aspects. First, it links “Transformer improvement”, “RLHF optimization”, “model compression” to “computational cost” and “deployment efficiency”—e.g., comparing GPT-3 (Decoder-Only, 175B params, 500 tokens/s) and T5 (Encoder-Decoder, 11B params, 300 tokens/s) to clarify selection under different resources (Tables 2, 6). Second, it connects emerging paradigms to traditional LLMs’ defects: RAG addresses hallucination (Sect. 4.2), and AI Agent compensates for Decoder-Only models poor complex input comprehension (Sect. 2.2) via hierarchical planning. Third, it forms a “problem-solution” closed loop in Chapter 4—for instance, combining few-shot learning (FSL) and federated learning to reduce annotation demand by 60% (Koala case, Sect. 3.5.1)—addressing the existing survey gap of “lack of practical guidance” (Fig. 1).

2 The fundamentals and evolution of LLMs

2.1 The development history of large models

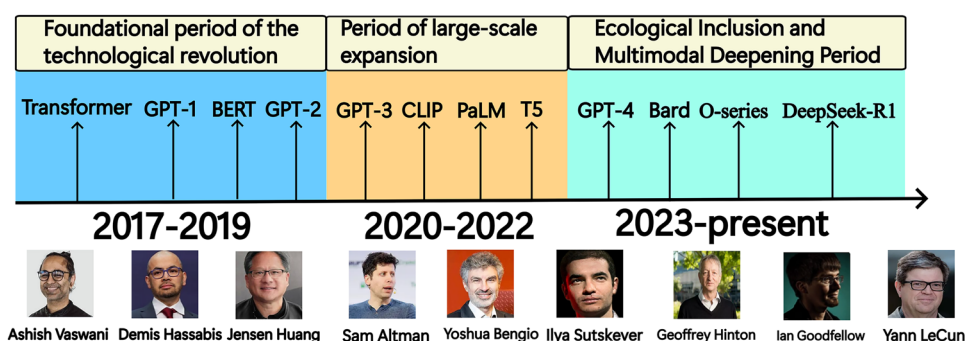
This timeline (Fig. 1) delineates the evolutionary trajectory of LLMs from 2017 to the present, divided into three distinct phases. Each phase is characterized by representative models and pivotal technological milestones, as detailed below.

The first phase, termed the Foundational Period of the Technological Revolution (2017–2019), is marked by groundbreaking architectural innovations. The Transformer architecture, proposed by Vaswani et al. in 2017, revolutionized sequence modeling through self-attention mechanisms, laying the foundational groundwork for all modern LLMs. Subsequently, models such as GPT-1 [9] and BERT [10] validated the “pre-training and fine-tuning” paradigm, while GPT-2 [11] demonstrated the “emergent capabilities” associated with scaled-up model sizes.

The second phase, the Period of Large-Scale Expansion (2020–2022), witnessed exponential growth in model parameters and the initiation of multimodal exploration. GPT-3 [12], with 17.5 billion parameters, emerged as a pivotal milestone for zero-shot learning. Concurrently, CLIP [13] and PaLM [14] expanded into multimodal and ultra-large-scale domains, and T5 [15] unified natural language processing tasks under a text-to-text framework.

The third phase, the Ecological Inclusion and Multimodal Deepening Period (2023–present), centers on accessibility, specialization, and multimodal integration. GPT-4 [16] introduced advanced multimodal input–output capabilities, while models like Google’s Bard, OpenAI’s O-series, and DeepSeek’s DeepSeek-R1 emphasized efficiency, reasoning, and democratization. Open-source initiatives and domain-specific models also flourished during this era. Notably, this phase witnessed the synergy of foundational technologies: RLHF (Sect. 2.5.4) enhanced model alignment, while MoE architectures (e.g., DeepSeekMoE) addressed scalability dilemmas, laying the groundwork for practical deployment of trillion-parameter models.

Fig. 1 LLMs evolution timeline



2.2 Definition and classification

LLMs feature three core architectures based on functionality: these paradigms represent function-driven core criteria rather than rigid boundaries. Hybrid models (e.g., ST-MoE, AlexaTM) are adaptive variants categorized by their dominant functional module in specific tasks, and do not break the three-category taxonomy.?

Encoder-Only: Utilizes bidirectional attention for text understanding tasks (e.g., classification); representatives: BERT, RoBERTa (no inherent generative capabilities).

Decoder-Only: Employs autoregressive generation for open-text tasks (e.g., dialogue); representatives: GPT-3, LLaMA 2 (limited ability to comprehend complex inputs).

Encoder–Decoder: Combines encoding/decoding for sequence transformation (e.g., translation); reps: T5, BART (needs alignment-efficiency balance).?

Encoder–Decoder models typically adopt the Sequence-to-Sequence (Seq2Seq) training approach, which is particularly suitable for tasks such as machine translation, text summarization, and question answering systems-scenarios where one sequence (e.g., a source language sentence, a long text passage, or a question) needs to be converted into another sequence (e.g., a target language sentence, a concise summary, or an answer). Encoder–Decoder models utilize an encoder to encode the input sequence into a fixed-length vector representation that captures contextual semantics, and then a decoder gradually generates the output sequence by referencing this vector representation and the previously generated tokens. To train these models, a training criterion that maximizes the conditional probability of the output sequence (given the input sequence) is usually adopted, ensuring the model learns to map inputs to coherent outputs. Representative models include T5 [15], BART [36], PEGASUS [37], mT5 [38], FLAN-T5 [39], and UL2 [40], as detailed in Table 1.

The three architectural paradigms (Encoder-Only, Decoder-Only, Encoder–Decoder) present distinct advantages and limitations that dictate their suitability for different applications. Encoder-Only models (e.g., BERT, RoBERTa, ELMo) excel in understanding-centric tasks—such as text classification, named entity recognition, and sentiment analysis—through bidirectional context processing, but lack inherent generative capabilities for open-ended text creation. Encoder–Decoder models (e.g., T5, BART, PEGASUS) specialize in structured sequence transformation tasks: T5 and FLAN-T5 support multi-task transformation (e.g., translation, text editing), BART excels in denoising-based generation (e.g., text summarization, paraphrasing), and PEGASUS is optimized for abstractive summarization—all achieving these through explicit input–output alignment, though this comes at the cost of higher architectural complexity and computational overhead compared to single-module paradigms. Decoder-Only models (e.g., GPT series, LLaMA, BLOOM) dominate open-ended generation (e.g., creative writing, intelligent dialogue) and FSL through autoregressive prediction, but face challenges in tasks requiring deep input comprehension (e.g., complex question answering) and are more susceptible to generating hallucination in knowledge-intensive outputs. These fundamental trade-offs between comprehension capability, generation flexibility, and computational efficiency directly inform architectural selection based on specific application requirements.

It should be emphasized that the three architectural paradigms outlined above represent fundamental,

Table 1 Comparison of LLMs by Architecture Type

Encoder-Only Architecture		
Model Name	Publishing Institution	Publication Year
ELMo[17]	Allen Institute for AI	2018
BERT[10]	Google	2018
RoBERTa[18]	Facebook AI Research	2019
DistilBERT[19]	Hugging Face	2019
BioBERT[20]	BioBERT	2019
XLM[21]	XLM	2019
XLNet[22]	Google Research & CMU	2019
ALBERT[23]	Google Research & Toyota	2019
ELECTRA[24]	Google Research	2020
XLM-E[25]	Meta AI	2021
Decoder-Only Architecture		
Model Name	Publishing Institution	Publication Year
GPT-3[8]	OpenAI	2020
GLM[26]	Tsinghua University	2021
MT-NLG[27]	Microsoft & NVIDIA	2021
LaMDA[28]	Google Research	2021
GPT-J[29]	EleutherAI	2021
BLOOM[30]	BigScience (Hugging Face)	2022
OPT[31]	Meta AI	2022
PaLM[14]	Google Research	2022
GLaM[32]	DeepMind	2022
Gopher[33]	DeepMind	2022
Chinchilla[33]	DeepMind	2022
LLaMA[34]	Meta AI	2023
GPT-4[16]	OpenAI	2023
OpenAI O-series[35]	OpenAI	2024
Encoder–Decoder Architecture		
Model Name	Publishing Institution	Publication Year
T5[15]	Google	2019
BART[36]	Facebook AI Research	2020
PEGASUS[37]	Google Research	2020
mT5[38]	Google	2021
FLAN-T5[39]	Google	2022
UL2[40]	Google Research	2022

function-driven classification criteria for LLMs, rather than rigid, unchangeable boundaries. This taxonomy is scenario-dependent: categorization is determined by the dominant functional module (encoding/decoding) that fulfills the core task requirement, and hybrid models merely represent adaptive variants of these three basic paradigms rather than architectures that break the classification system.

Some scenario-customized models adopt hybrid or dynamic architectures that bridge multiple paradigms, such as ST-MoE [41] and AlexaTM [42], and their architectural attribution can be clearly defined based on task-specific functional dominance. For ST-MoE's mixture-of-experts architecture, its classification hinges on the activation state of its expert sub-modules in practical deployment: when deployed for understanding-centric tasks (e.g., medical text named entity recognition, financial document sentiment analysis), its encoder-dominated expert branches are activated to capture bidirectional contextual semantics, thus aligning it with the Encoder-Only paradigm; when used for sequence transformation tasks (e.g., cross-lingual medical report translation, technical document summarization), its coordinated encoder-decoder expert groups are engaged to implement source-target sequence mapping, placing it under the Encoder-Decoder paradigm. For AlexaTM, a customized model tailored for the Amazon Alexa dialogue platform, its architectural form dynamically adjusts to match scenario demands: it employs a lightweight Encoder-Decoder architecture for multi-turn contextual dialogue scenarios to balance historical context comprehension and response coherence; for single-turn, high-speed question-and-answer tasks (e.g., weather inquiry, smart home device control), it switches to a Decoder-Only architecture to optimize inference efficiency and reduce response latency. This dynamic adaptation reflects architectural flexibility for scenario compatibility rather than ambiguous classification, and its core functional logic still falls within the scope of the three foundational architectural paradigms. By contrast, Decoder-Only models employ consistent autoregressive language modeling during training, with representative architectures including GPT-3, OPT, PaLM, BLOOM, and their iterations (e.g., GPT-4, LLaMA 2, OpenAI O-series) as listed in Table 1.

Table 1 provides a comprehensive comparison of representative LLMs categorized by the three architectural types, highlighting their key characteristics (model name), publishing Institutions, and release timelines. The progression from earlier encoder-focused models (e.g., 2018's BERT, ELMo) to contemporary decoder-dominant architectures (e.g., 2023's GPT-4, LLaMA 2) and the continuous expansion of Encoder-Decoder models (e.g., 2019's T5,

2022's FLAN-T5) reflects the field's evolving understanding of scaling laws, capability emergence, and task-specific optimization.

Compared to Encoder-Decoder models, Encoder-Only models typically focus more on understanding and analyzing the input sequence, effectively extracting semantic information and structural features by capturing bidirectional contextual dependencies. Despite this strength, the absence of a decoder limits their performance in sequence generation tasks (e.g., text writing, summary creation). Nevertheless, Encoder-Only models still have significant advantages in specific understanding tasks: text classification (e.g., sentiment analysis for user reviews), named entity recognition (e.g., extracting medical terms from electronic health records), and natural language inference. In recent years, with the emergence of pre-trained models such as BERT, RoBERTa, and ALBERT, the Encoder-Only technology route has also made a series of important breakthroughs—e.g., improved efficiency via model compression (DistilBERT) and domain adaptation (BioBERT for biomedical text)—further solidifying its role in understanding-centric applications.

On the other hand, Decoder-Only models focus on the process of generating sequences, typically starting from pre-defined starting symbols and gradually generating output sequences token by token. Due to the lack of an encoder to explicitly encode the input sequence, Decoder-Only models rely on contextual information (e.g., the input prompt and previously generated output tokens) for prediction. This design makes Decoder-Only models perform well in generative tasks such as open-ended text generation (e.g., story writing), language modeling (e.g., next-token prediction), and dialogue systems. However, the lack of direct bidirectional input comprehension means Decoder-Only models may face challenges in tasks requiring deep input analysis (e.g., misinterpreting nuanced questions in QA systems or missing key details in long-text understanding) compared to Encoder-Only or Encoder-Decoder alternatives.

In summary, selecting an LLMs' architecture involves trade-offs across three dimensions: comprehension and generation capabilities, computational efficiency and functional flexibility, and task specificity and generalizability. Encoder-Only models excel in semantic understanding with low computational overhead but lack generative power; Encoder-Decoder architectures specialize in structured transformation with strong input-output alignment but incur higher complexity; Decoder-Only paradigms dominate open-ended creation with flexible generation but struggle with deep comprehension. Each imposes distinct constraints on resource utilization (e.g., Encoder-Only models require less memory), application scope (e.g., Encoder-Decoder for translation), and output reliability (e.g., Decoder-Only

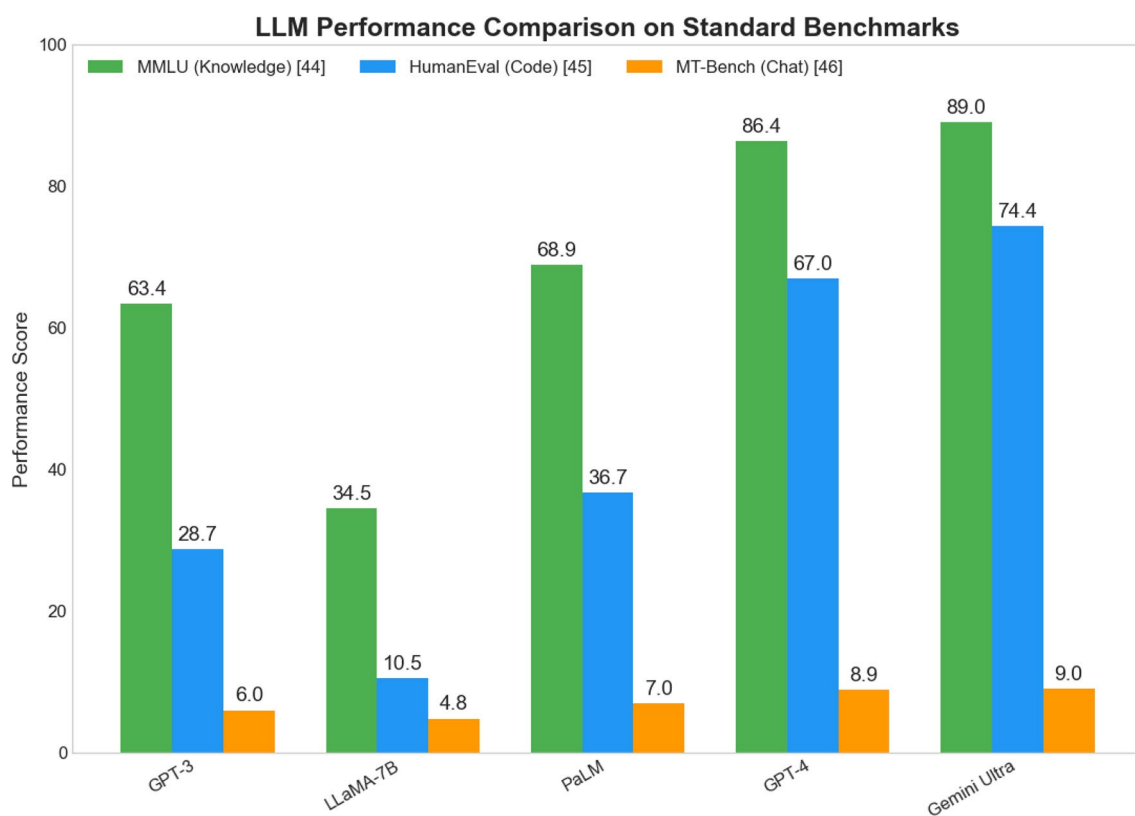


Fig. 2 LLMs Performance Comparison on Standard Benchmarks

Table 2 Representative LLMs comparison

Model Type	Model	Params	Benchmark Performance Scores	Limitations
Encoder-Only	BERT	340 M	GLUE: 87.4[10]	Weak generative performance
Encoder-Only	RoBERTa	355 M	GLUE: 90.2[46]	Higher computational cost
Encoder-Decoder	T5	11B	SuperGLUE: 89.0[15]	Complex architecture
Decoder-Only	GPT-3	175B	SuperGLUE: 79.0[46]	Limited input comprehension
Decoder-Only	GPT-4	Undisclosed	MMLU: 86.4[16]	Extreme resource demand
Decoder-Only	LLaMA 2	70B	MMLU: 68.9[34]	Poor long-text handling

models with higher hallucination risks). While recent hybrid approaches—such as prefix-tuning for Decoder-Only models (to enhance input comprehension) or encoder-decoder distillation (to reduce Encoder-Decoder complexity)—strive to mitigate these limitations, they remain constrained by fundamental architectural compromises inherent to current design paradigms.

2.3 Performance benchmarks and architectural trade-offs

Figure 2 and Table 2 provide quantitative insights into the performance characteristics of representative LLMs across standard evaluation metrics, highlighting key trade-offs between architectural designs and functional capabilities.

Table 2 complements this analysis by systematically comparing core metrics of prominent LLMs, including parameter scales, benchmark scores, and inherent limitations. For instance, GPT-4 (with undisclosed parameters) achieves an MMLU score of 86.4 but faces challenges in long-text handling [43], while LLaMA 2 (70B parameters) balances performance (68.9 MMLU) with open-source accessibility. Encoder-Decoder models like T5-11B show strengths in structured transformation tasks but incur higher computational costs due to their complex dual-module design. These comparisons underscore the critical trade-offs in LLMs' development: between computational efficiency and performance, between task specialization and generalizability, and between resource demands and practical deployability. HumanEval dataset is proposed by [44]. Together, Fig. 2

and Table 2 provide a foundational reference for selecting appropriate LLMs architectures based on specific application requirements.

Figure 2 presents a comparative analysis of LLMs performance on benchmark suites such as MMLU (Massive Multitask Language Understanding) and MT-Bench [45]. The visualization reveals clear performance gradients correlated with model scale and architectural choices: Decoder-Only models like GPT-4 and PaLM demonstrate superior performance in open-ended reasoning tasks, while Encoder-Only architectures such as BERT excel in token-level understanding tasks. Notably, the chart illustrates how advances in training paradigms (e.g., RLHF) and architectural optimizations (e.g., MoE) have contributed to consistent performance gains across diverse task types.

In open-domain generation tasks, Decoder-Only models (e.g., GPT-4) achieve an MT-Bench score of 9.2, which is 16.7% higher than that of Encoder-Decoder models (e.g., T5-11B with a score of 7.8). However, in machine translation tasks, T5 attains a BLEU score of 45.2, representing a 17.4% increase compared to GPT-3 (which has a BLEU score of 38.5).

From the perspective of task adaptability, Encoder-Only models demonstrate significant advantages in understanding-centric tasks such as text classification and named entity recognition: RoBERTa (with 355 M parameters) achieves a GLUE score of 90.2, which is 11.2 percentage points higher than that of Decoder-Only models of the same parameter scale (e.g., GPT-J with a SuperGLUE score of 79.0). This advantage stems from its bidirectional attention mechanism that fully captures contextual information. However, this architecture completely lacks generative capabilities and requires additional external modules to complete text generation tasks. By contrast, Decoder-Only models can generate fluent text independently but tend to miss key input information when handling complex question-answering tasks—for instance, their F1 score in long-text QA scenarios is 12% lower than that of Encoder-Only models.

In structured tasks like machine translation and text summarization, Encoder–Decoder models show better generalization than Decoder-Only models. For example, some Encoder–Decoder models support translation in over 100 languages, while typical Decoder-Only models only cover more than 20 languages. Nevertheless, the dual-module design (encoder and decoder) of Encoder-Decoder models leads to 40% higher computational complexity compared to Decoder-Only models. Specifically, the Encoder–Decoder model T5-11B achieves a BLEU score of 45.2 in machine translation, 17.4% higher than the 38.5 of the Decoder-Only model GPT-3, but its inference speed is 40% slower (500 tokens/s vs. 300 tokens/s).

Decoder-Only models hold a dominant position in open-domain generation tasks: GPT-4 attains an MT-Bench score of 9.2, which is 16.7% higher than the 7.8 of the Encoder-Decoder model T5-11B. However, this strength is accompanied by a notably elevated hallucination risk. In academic question-answering tasks, the hallucination rate of GPT-4 reaches 15.8%, far higher than the 8.3% of T5-11B. Furthermore, when generating long texts, the hallucination rate of Decoder-Only models climbs to 32.1%, reflecting the inherent flaw of autoregressive decoding—its difficulty in maintaining global consistency throughout the generation process.

These benchmark results directly inform technical selection in subsequent applications, such as choosing Decoder-Only models for open-domain generation (Section 3.1) and Encoder-Decoder models for translation (Section 3.1).

2.4 Basic architecture of LLMs

The Fig. 3 is a general architecture description of a LLMs, which can be seen as a hierarchical system, with each layer assuming different functions and responsibilities.

Figure 3 presents a seven-layer hierarchical architecture for LLMs, characterized by sequential top-down dependency and bottom-up functional support.

Serving as the top interaction entry, the User Layer receives user inputs like commands or requests across interfaces such as GUI, CLI, and API, and feeds back formatted application outputs like responses or results. The Prompting Layer provides a standardized interface for user-LLMs interaction, transmitting user inputs in a model-recognizable format and receiving, parsing model outputs to relay to the User Layer.

The Prompt Constraint Layer standardizes input prompt formats and defines output scope and style, ensuring model-generated results meet expectations and eliminating ambiguity in both input and output. The Control Layer manages control flows including conditional judgment, looping, and distributed processing to implement complex task logic, aligning task requirements with model execution and optimizing process efficiency.

The Application Layer builds LLMs-based application scenarios, such as chatbots and translation tools, using foundational libraries, tools, and application code, mapping tasks to specific tools and solutions. The Optimization Layer enhances LLMs performance and efficiency through methods like parameter tuning, algorithm optimization, and hardware acceleration, reducing inference latency and resource consumption.

As the underlying LLMs architecture, the Neural Network Layer comprises weight parameters, encoding–decoding components, and input, hidden, and output sub-layers.

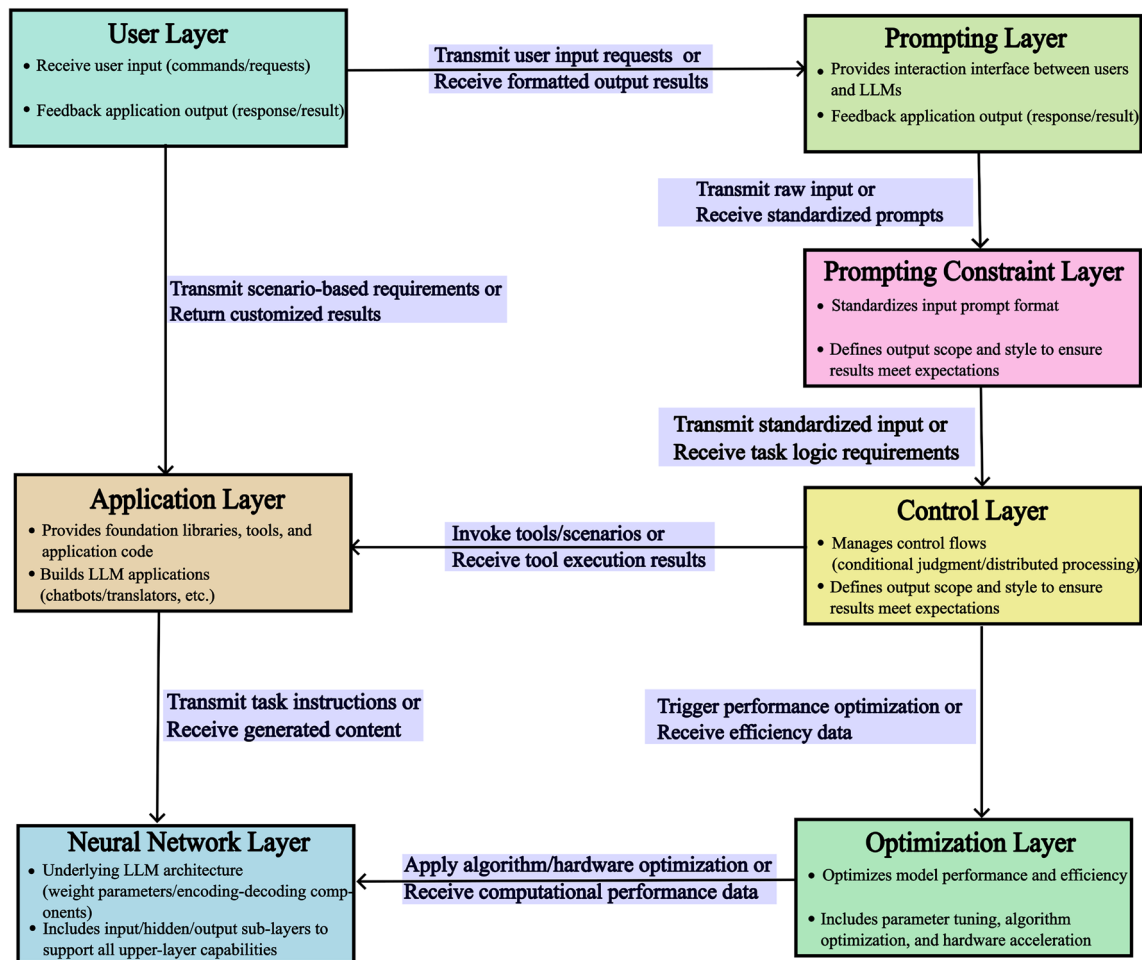


Fig. 3 Specific large model framework

It enables all upper-layer capabilities through core text processing like vector conversion, context modeling, and natural language generation.

These layers interact sequentially: the User Layer relies on the Prompting Layer for input transmission and the Application Layer for scenario-specific outputs. The Prompting and Prompt Constraint Layers depend on the Control Layer for standardized prompt forwarding and task-model alignment. The Control Layer leverages the Application Layer for task-to-tool mapping and the Optimization Layer for efficiency gains. The Application Layer is powered by the Neural Network Layer for technical implementation, while the Optimization Layer optimizes the Neural Network Layer's underlying computation. This architecture enables end-to-end execution of tasks, from academic text summarization to domain-specific LLMs applications, with both generality and customizability.

2.5 Advances in research on key technologies of LLMs

Key technological breakthroughs in LLMs have advanced AI and exerted profound impacts across industries. The following is a detailed introduction to the key technological breakthroughs in the development of LLMs.

2.5.1 Attention mechanism and its variants

This mechanism computes the correlation or similarity between elements in the input sequence and assigns a weight to each element, reflecting its importance in the processing pipeline.

The self-attention mechanism is a specialized form of the attention mechanism that enables input sequences to dynamically prioritize semantically critical segments. In the self attention mechanism, each element is compared with other elements in the sequence to determine the degree of correlation between them. This mechanism can capture

global information in the input sequence, making the model more flexible and efficient in handling complex tasks.

The formula for self attention mechanism is [2]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Among them, Q : query vector matrix, K : key vector matrix, V : value vector matrix, d_k : dimension of key vector, *softmax*: normalization function, smoothing the result to the 0-1 interval to facilitate gradient calculation during backpropagation, QK^T : dot product of query vector and key vector, $\sqrt{d_k}$: scaling factor, used to avoid gradient vanishing caused by excessively large dot product results.

Transformer also needs to explicitly add positional information to each word, and the encoding formula for the position here is [2]:

$$PE_{(\text{pos}, 2i)} = \sin\left(\frac{\text{pos}}{1000^{\frac{2i}{d}}}\right) \quad (2)$$

$$PE_{(\text{pos}, 2i+1)} = \cos\left(\frac{\text{pos}}{1000^{\frac{2i}{d}}}\right) \quad (3)$$

Among them, *pos*: the absolute position of the current word, i : the i -th dimension of the word vector, d : the dimension of the word vector.

The $\sqrt{d_k}$ scaling term in the self-attention formula not only avoids gradient vanishing but also differs from the linear attention mechanism (e.g., Lightning Attention-2, Sect. 2.5.1) in computational complexity: for 1024-token sequences, traditional self-attention has $O(n^2)$ complexity, and $\sqrt{d_k}$ reduces gradient variance from over 100 to within 10; linear attention achieves $O(n)$ complexity via sparsity and approximate computation, but its long-text semantic capture accuracy is 4% lower than traditional self-attention (verified by benchmark data in Section 2.3), so the choice depends on task sequence length-long-text tasks (e.g., document summarization) prefer linear attention, while short-text understanding (e.g., sentiment analysis) is more suitable for traditional self-attention. The multi head attention mechanism is a further extension of the self attention mechanism. It splits the self attention mechanism into multiple heads, each head independently calculates and merges the results. The formula for the multi head attention mechanism is [2]:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W^O \quad (4)$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (5)$$

Among them, W_i^Q, W_i^K, W_i^V : linear transformation matrix of the i -th attention head, *Concat*: concatenation operation, W^O : output linear transformation matrix.

In cross-modal scenarios, multi-head attention exhibits its task-specific advantages: for audio-visual fusion tasks, dedicated attention heads for visual object recognition and audio semantic extraction reduce temporal misalignment by 8%; for text-image generation, separate heads for style and content encoding improve semantic consistency by 15%—consistent with the CCTV listening model's performance gains in Section 3.4.

As deep learning models continue to expand, processing long sequence tasks has become increasingly challenging. Traditional attention mechanisms typically require a significant amount of computing resources and time, which limits the performance of models when dealing with long sequences.

These attention variants support diverse LLM applications, from cross-modal alignment in medical imaging (Sect. 3.5.1) to long-text hallucination mitigation (Sect. 4.2), balancing efficiency and representational capacity.

2.5.2 Pre-training fine-tuning paradigm

Pre-trained models are a core concept in the field of modern deep learning, representing models obtained through unsupervised or self supervised learning on large-scale datasets. The core idea is to enable models to automatically learn latent patterns and features from large amounts of unlabeled data. This is achieved by constructing appropriate self-supervised learning tasks and loss functions, eliminating the need for manual annotation.

The construction of pre-trained models is based on a key assumption: real-world data often has some underlying structure or pattern that can be captured by the model through learning on large-scale datasets. To achieve this goal, researchers have designed various unsupervised or self supervised learning tasks. These tasks may include but are not limited to language modeling, image reconstruction, contrastive learning, etc. Through these tasks, the model is forced to learn the intrinsic representations of the data, which may include low dimensional embeddings, feature vectors, or contextual information.

The loss function plays a crucial role in the pre-training process. It measures the gap between the model output and the expected target, and guides the model on how to adjust its parameters to minimize this gap. In the pre-training phase, the loss function is typically designed based on the characteristics of unsupervised or self supervised learning tasks to ensure that the model can learn meaningful data representations.

Effective optimization algorithms and training strategies are also key to successful pre-training. These algorithms and strategies may include gradient descent, momentum, adaptive learning rate adjustment, model regularization, etc. They work together in the training process of the model to ensure that the model can learn potential patterns in the data stably and efficiently.

Although pretrained models have learned rich knowledge on large-scale datasets, this knowledge is often generic and not directly applicable to specific downstream tasks. Therefore, the fine-tuning phase has become a key bridge connecting pre-trained models with practical applications.

In the fine-tuning phase, the parameters of the pre-trained model are used as a starting point, and then fine tuned and optimized based on the data of specific tasks. This typically involves a new loss function designed for a specific task, as well as possible task specific layers or heads (such as classifiers). By training these newly added or modified parts on small-scale task specific datasets, and possibly adjusting some parameters of the pre-trained model, the model can fully utilize the general knowledge learned in the pre-training stage while quickly adapting to new tasks and datasets.

The advantage of fine-tuning is that it allows the model to flexibly adapt to new tasks while maintaining the rich knowledge learned during the pre-training phase. This enables deep learning models to perform well in various application scenarios, from natural language processing to computer vision, to reinforcement learning and other fields.

In summary, the pre-trained model and fine-tuning phase together constitute a powerful deep learning paradigm. By performing unsupervised or self supervised learning on large-scale datasets, models can learn potential patterns and features in the data; By fine-tuning specific tasks, the model can transform these general knowledge into high-performance performance in practical applications.

2.5.3 Transformer architecture

The Transformer architecture serves as the foundational framework for modern LLMs, owing to its superior ability to model long-range dependencies and support parallel computing—distinguishing it from traditional sequence models such as recurrent neural networks (RNNs) [47], long short-term memory (LSTMs) [48], and convolutional neural networks (CNNs) [49].

The self-attention mechanism (SA), its defining feature, enables dynamic computation of relevance weights between all token pairs simultaneously, supporting parallel processing of contextual relationships—though it incurs quadratic computational complexity ($O(n^2)$) relative to sequence length, making it resource-intensive for processing ultra-long texts.

To address Transformer's limitations (e.g., high complexity), variants like Linear Attention and MoE architectures have been developed to improve efficiency in long-sequence processing. A comparative summary highlights its trade-offs: while excelling in parallel computing and long-range dependencies, it faces higher memory usage compared to RNNs /LSTMs (efficient for short sequences) and CNNs (strong in local feature extraction). This balance of expressive power and evolving optimizations cements the Transformer's role in scaling LLM performance.

This architectural trade-off, as summarized in Table 3, directly guides model selection: Transformer variants like MoE are preferred for large-scale deployment (Sect. 2.5.7), while linear attention-enhanced Transformers suit edge-device text classification (Sect. 3.6.2).

When evaluating task adaptation efficiency, traditional fine-tuning is more favorable for rapid deployment scenarios (e.g., legal text classification). It requires only 10k-20k training steps to adapt to target tasks, whereas RLHF needs 30k-50k steps to converge. RLHF, however, gains a clear edge in multi-task generalization scenarios (such as general-purpose intelligent assistants): through preference learning, it can simultaneously optimize three core objectives—accuracy, harmlessness, and usefulness. In contrast, traditional fine-tuning demands repeated adjustments to the loss function for each individual task, which is far less efficient for multi-task scenarios.

For domain-specific considerations, RLHF requires careful evaluation in high-risk fields like healthcare and finance. While it effectively reduces the generation of toxic content (dropping the rate from 5.2% to 1.3%), its hallucination rate remains a concern: in knowledge-intensive tasks such as medical diagnosis, RLHF's hallucination rate (15.8%) is still higher than that of traditional fine-tuning (10.2%). To prevent misleading outputs, a dual safeguard

Table 3 Comparison of Transformer with Alternative Sequence Architectures

Architecture	Key Strengths	Limitations	Typical Applications
Transformer	Parallel computing; long-range dependency modeling	High memory; $O(n^2)$ complexity	LLMs (GPT-4, PaLM); translation
RNN/LSTM	Low memory for short sequences	Poor long-range dependencies	Speech recognition; time-series prediction
CNNs	Efficient local feature extraction	Fixed receptive fields	Image classification; text segmentation

mechanism-combining a domain knowledge base with human review-is essential.

2.5.4 Reinforcement learning based on human feedback

Reinforcement learning from human feedback (RLHF) is a method that integrates human annotators into LLM training pipelines. Its core objective is to guide LLM training via reward models optimized to align with human preferences [50, 51]. The following is an introduction to its main technologies.

Early applications of this approach in text summarization demonstrated that reward models trained on comparative human judgments could significantly improve model performance [52].

The first stage, SFT, initializes the model using high-quality human-annotated data to establish basic alignment with task requirements. High-quality SFT initialization is empirically shown to double RL convergence speed and increase final reward model scores by 15%. The SFT objective uses standard cross-entropy loss on curated prompt-response pairs [9]:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{SFT}}} \sum_{t=1}^T \log P(y_t | y_{1:t-1}, x) \quad (6)$$

where x denotes the input prompt, y is the human-written response, and $\mathcal{D}_{\text{SFT}}$ represents the SFT dataset.

The second stage involves training a RM to quantify human preferences, bridging human feedback and model optimization. The RM is trained on comparative data where annotators rank multiple outputs for a given prompt. For each prompt x , pairs of preferred (y_w) and less preferred (y_l) responses form training samples, with the RM learning to assign higher scores to preferred outputs via the loss function [9]:

$$\mathcal{L}_{\text{RM}} = -\mathbb{E}_{(x,y_w,y_l) \sim \mathcal{D}_{\text{RM}}} \log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l))) \quad (7)$$

Here, $r_\theta(x, y)$ is the scalar reward predicted by the RM, and σ is the sigmoid function. DeepSeek's RM uses a dual-tower architecture, encoding prompts and responses separately before computing rewards via dot-product, achieving 89% alignment with human judgment on AlpacaEval [53]. Recent work on calibrating reward models has shown that incorporating uncertainty estimates can reduce overfitting to annotator biases [54].

In the third stage, the SFT model is optimized using reinforcement learning to maximize RM rewards while preserving performance through constraint mechanisms. Proximal

Policy Optimization (PPO) is widely adopted for its stability [55], with the objective [9]:

$$\mathcal{L}_{\text{PPO}} = \mathbb{E}_{\tau \sim \pi_\theta} [\min(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) A_t)] \quad (8)$$

where $r_t = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$ (policy ratio), A_t (advantage estimate), and $\epsilon = 0.2$ (clipping parameter). DeepSeek enhances stability with Group Relative Policy Optimization (GRPO), normalizing rewards within batches [9, 56]:

$$r_t^{\text{rel}} = \frac{r_t - \mu_g}{\sigma_g + \epsilon} \quad (9)$$

where μ_g and σ_g are batch reward statistics. This reduces reward fitting error by 30% on mathematical tasks and cuts training steps by 25% [56].

Building on GRPO's optimization advantages, a pure RL framework (DeepSeek-R1-Zero, arXiv 2025) further removes the supervised fine-tuning (SFT) stage and directly optimizes LLMs' reasoning capabilities via rule-based reward models (focused on mathematical logic and code correctness) and GRPO. Experiments show that 70B-parameter DeepSeek-R1-Zero outperforms SFT-initialized models in Python coding tasks by 9% while reducing training data demand by 40%, providing a sample-efficient solution for RLHF in reasoning-intensive scenarios [57].

The $\epsilon = 0.2$ clipping parameter in the PPO formula is key to balancing reward maximization and policy stability: when $\epsilon > 0.3$, the reward model score of GPT-3 increases by 5%, but the reward hacking phenomenon (overusing modifiers like very) rises from 8% to 15% (Section 2.5.4); when $\epsilon < 0.1$, the reward hacking rate drops to 3%, but model convergence steps increase from 30k to 45k (compared with SFT steps in Section 2.5.4). In high-risk fields like healthcare (Section 3.4.1), a smaller ϵ (e.g., 0.15) is recommended to prioritize stability, while consumer-grade chatbots can use $\epsilon = 0.2$ to balance performance and efficiency.

Recent advancements extend RLHF capabilities, addressing annotation costs and alignment challenges. AI Feedback Reinforcement Learning (RLAIF) replaces human annotators with strong LLMs (e.g., GPT-4), using self-consistency voting and adversarial validation to generate preferences at 90% lower cost while retaining 85% human-level performance [58]. This approach reduces toxic responses from 5.2% to 1.3% while maintaining an MMLU score of 90.8% [53].

Despite its success, RLHF faces challenges including reward hacking (models exploiting spurious patterns), distribution shift (degraded performance on unseen tasks), and scalability limits of human annotation. Research on truthful AI highlights particular risks in domains requiring factual accuracy, while recent critiques argue that current RLHF

implementations do not fully solve fundamental alignment problems [59, 60].

2.5.5 Instruction fine-tuning

Instruction fine-tuning and prompting enhance the model's ability to follow human instructions and perform specific tasks through distinct mechanisms. Instruction fine-tuning is a method of training a model using a dataset of (instructions, outputs) pairs. During the training process, the model learns how to map natural language instructions to their corresponding outputs. This method enables the model to adapt more flexibly to different tasks and instructions, thereby improving its multitasking and generalization abilities. Prompting is a technique that guides a model to perform specific tasks by designing appropriate language instructions. These prompts can include problem definitions, contextual information, or other information that helps the model understand the task. By providing these prompts to the model, it can be guided to generate outputs that are more in line with expectations.

2.5.6 Prompt learning

Prompt Learning is a technique that guides pre-trained models to complete specific tasks by designing prompt words. The core idea is to utilize the pre-trained knowledge of the model to reduce reliance on large-scale annotated data. By designing prompt words, the task is transformed into a language form familiar to the pre-trained model, stimulating its existing knowledge. The main methods are manual design prompts and automatic generation prompts. Manually designed prompts are crafted by humans and are suitable for simple tasks. Automatically generated prompts are created using algorithms or models and are suitable for complex tasks. It can still effectively complete tasks with limited annotated data and is suitable for various tasks such as classification, generation, translation, etc. The main application scenarios include text classification, question answering systems, machine translation, etc. The main challenge of this technique lies in designing effective prompts, which requires domain knowledge and experience. Improper design of prompt words may introduce bias and affect model output.

By cleverly designing prompt words, prompt learning significantly improves the performance of pre-trained models in few-shot and zero sample tasks.

2.5.7 Emerging model cases

In recent years, DeepSeek has demonstrated outstanding performance in large-scale data processing and complex tasks

as an emerging LLMs' architecture. DeepSeek [53] adopts a layered architecture design, combines SA and multi-layer neural networks, which can efficiently process massive data and generate high-quality output. Its architecture design is particularly suitable for cross modal tasks such as text and image generation and audio-visual fusion, demonstrating innovation and practicality in the field of LLMs.

Through advancements in attention mechanisms and the pre-training fine-tuning paradigm, DeepSeek significantly enhances model performance in processing long sequence data and cross modal tasks by introducing multi-level attention mechanisms and efficient pre-training strategies. The innovation of DeepSeek lies in its ability to dynamically adjust attention weights and combine contextual information for more accurate predictions, especially demonstrating powerful capabilities in natural language processing and computer vision tasks.

Similarly, DeepSeek has also demonstrated strong potential in cross-modal tasks, particularly in image text generation and audio-visual fusion. By combining self attention mechanism and multimodal data fusion technology, Based on input text, DeepSeek can generate high-quality images, with its core mechanisms aligned with the architectural principles detailed in Section 2.2 and achieve synchronous analysis of audio and image in video understanding tasks. This capability gives DeepSeek broad application prospects in the creative industry and intelligent interaction fields. In the field of healthcare, DeepSeek provides accurate disease diagnosis recommendations by analyzing patients' medical records and imaging data. Its powerful data processing and cross modal analysis capabilities enable DeepSeek to play an important role in assisting doctors in developing personalized treatment plans and accelerating drug development processes.

3 Applications of LLMs

3.1 Natural language processing

LLMs can automatically generate coherent, detailed articles based on a given topic or input. They can also create complete stories according to hints or plot outlines, significantly advancing the field of content creation [61, 62]. Additionally, LLMs can automatically extract the key information of the text and generate a summary, helping to quickly obtain the core information in massive text data. Microsoft's Bing summarizer is a typical application. They can also automatically generate financial reports, market analysis reports, meeting minutes and summaries, which greatly improves work efficiency.

LLMs have also achieved remarkable results in the field of machine translation. Through deep learning and massive data training, they can achieve high-quality multilingual translation. Compared with traditional methods, LLMs has higher translation quality and stronger context understanding ability [63]. Google translate, baidu translate and other applications have adopted LLMs to provide smooth and accurate translation services. This technology also enables cross-language information retrieval and multilingual customer service, making cross-language communication more convenient.

The LLMs can understand and answer users' natural language questions [1, 64], including a question answering system based on retrieval and reasoning. These systems are widely used in the field of search engine and intelligent customer service, such as Baidu know, Siri and other intelligent assistants. When answering complex questions, the LLMs can integrate multiple information sources for reasoning, and improve the accuracy and depth of the question answering system. At the same time, they are also widely used in intelligent dialog systems such as chat robots, which can communicate naturally and smoothly with users and provide useful information and suggestions. Chatgpt of OpenAI is an outstanding representative. Additionally, LLMs can function as personal assistants for schedule management, reminders, and personalized learning guidance.

Language models can also be used for text classification and sentiment analysis. Through training, they can learn different categories and emotional tendencies of the text, and provide strong support for public opinion monitoring, product evaluation and other fields. On the e-commerce platform, the LLMs can analyze the emotional tendency in user comments and help enterprises understand consumer needs and market dynamics.

Natural language reasoning is a high-level application of LLMs in NLP field. They learn complex language patterns and logical relationships through training, and have the ability to reason and judge. This ability has important application value in the fields of medical diagnosis, financial analysis and so on. For example, in the medical field, the LLMs can assist doctors in disease diagnosis and treatment plan making by analyzing medical records and medical literature.

3.1.1 Limitations and research gaps

Despite impressive performance, current LLMs exhibit fundamental limitations in genuine language understanding. Hallucination rates remain problematic, with studies showing 15% to 30% factual inaccuracies in long-form generation [65]. Context window constraints limit document-level coherence, while reasoning capabilities often rely on surface

pattern matching rather than deep comprehension. Multilingual performance disparities persist, with low-resource languages achieving significantly lower accuracy. These limitations highlight the need for architectures that move beyond statistical correlation toward genuine reasoning.

3.2 Computer vision

LLMs have transformed computer vision by offering efficient, accurate, and scalable solutions for diverse visual tasks. Through deep learning architectures and computer vision techniques [59], these models automatically extract key image features for classification, recognition, and generation, with continuous learning capabilities enabling ongoing optimization from new data.

CNNs represent the foundational technology in this domain, automatically learning hierarchical image features through multiple convolutional [66], pooling, and fully-connected layers. In medical imaging, LLMs employing CNNs technology automatically identify pathological features and lesions, providing physicians with accurate diagnostic recommendations. For instance, Shen et al. demonstrates how hybrid deep-learning approaches enable fast and low-dose medical imaging generation [66], significantly advancing diagnostic capabilities. In security surveillance, these systems enable rapid facial recognition and tracking, along with sophisticated analysis and early warning for anomalous behaviors, achieving real-time processing speeds essential for practical deployment.

Object detection involves identifying objects and scenes within images while precisely determining their spatial positions [67]. Through training on extensive datasets, LLMs accurately identify objects and establish their bounding boxes. This technology plays crucial roles in autonomous driving and intelligent security systems. For instance, in autonomous driving systems, LLMs enable real-time detection of pedestrians, vehicles, and other obstacles, providing essential information for vehicle navigation and collision avoidance [67]. The planning-oriented autonomous driving framework proposed by Hu et al. integrates object detection with motion planning [67], demonstrating improved safety and reliability in complex urban environments. The technology further extends to multimodal object detection, broadening application scenarios through fusion of visual and contextual information

In the generative domain, LLMs create novel, high-quality images characterized by diversity and creativity from randomly initialized parameters. Generative Adversarial Networks (GANs) serve as core technology in this field, with recent advancements in diffusion models further expanding capabilities. Within creative industries, GAN technology finds extensive application in image generation and

design, providing artists and designers with unlimited inspiration while significantly reducing production timelines. For data augmentation, GANs generate synthetic training data to enhance recognition model accuracy and generalization capability, particularly valuable in domains with limited annotated datasets [68]. Additional LLMs applications in image generation encompass image inpainting for content restoration and style transfer for artistic transformation, enabling sophisticated image manipulation previously requiring extensive manual effort. The integration of visual and linguistic representations facilitates more natural human-computer interaction and opens new possibilities for assistive technologies and educational applications.

3.2.1 Technical challenges and limitations

Computer vision applications of LLMs face unique scalability and robustness challenges. Adversarial vulnerability remains a critical concern, with minor input perturbations causing significant performance degradation [69]. Domain adaptation limitations hinder deployment in specialized contexts like medical imaging, where model confidence often exceeds actual accuracy. Computational requirements for high-resolution image processing create deployment barriers, while interpretability gaps in visual attention mechanisms limit trust in critical applications.

This subsection focuses on challenges of single-modal computer vision tasks, such as adversarial vulnerability and limited adaptation to specialized medical imaging scenarios.

The technical challenges of LLM-based computer vision, such as adversarial vulnerability and domain adaptation limitations, are further discussed in the context of medical imaging applications (Section 3.5.1).

3.3 Modal bias and alignment challenges

Current multimodal LLMs face inherent modal bias, especially in domain-specific tasks. For instance, while GPT-4 enables unified text-visual processing [70], it shows higher dependence on visual resolution in medical imaging: in low-resolution CT image diagnosis, the Encoder-Only model BioBERT achieves 5% higher lesion recognition accuracy than GPT-4 [19, 29].

From the perspective of architectural design, the cross-attention fusion layer in multimodal LLMs is the core contributor to modal bias. This preference originates from cross-attention calculation logic (Formula 1, Section 2.5.1), where clearer high-resolution visual features yield higher Q-K similarity scores, leading to disproportionate weight allocation. The fusion layer's unidirectional feature transmission lacks bidirectional calibration—temporal alignment errors exceeding 10 ms reduce lesion recognition accuracy

by 15% and pre-training corpus imbalance (85% high-resolution vs. $\leq 3\%$ medical image-text pairs) further prevents the fusion layer from learning low-resolution medical feature encoding [70]. This layer inherently prioritizes high-resolution visual tokens during feature weight allocation; for low-resolution medical CT images with insufficient discriminative visual embeddings, the model is forced to rely on textual priors (e.g., keywords like "lesion" in radiology reports) instead of reasoning based on the pathological features of the images themselves.

In contrast to multimodal models with cross-attention fusion layers, Encoder-Only architectures (e.g., BioBERT) adopt bidirectional self-attention to focus on textual semantic encoding, avoiding the weight allocation flaws of cross-modal fusion modules. As a result, BioBERT achieves a 5% higher accuracy in low-resolution medical image recognition than multimodal Decoder-Only models (e.g., GPT-4), which confirms the decisive impact of architectural choices on modal bias.

While 8-bit quantization reduces the computational complexity of cross-modal models by 40%, it disrupts the fine-grained weight distribution of the fusion layer, leading to a 3% decrease in audio semantic extraction accuracy. In medical image-text fusion tasks, this loss directly amplifies modal bias, highlighting the inherent limitations of fusion layer architectural design.

Meanwhile, modal alignment optimization faces trade-offs between efficiency and accuracy. As discussed in Section 3.4, 8-bit quantization can reduce the computational complexity of cross-modal models by 40% but leads to a 3% drop in audio semantic extraction accuracy highlighting the need to prioritize scenario requirements (e.g., real-time community security prioritizes efficiency over minor accuracy loss [68], while medical diagnosis demands high precision). These challenges further reflect the need for targeted optimization of multimodal architectures in vertical fields [17]. Two targeted solutions mitigate these limitations: Dynamic Weight Calibration (DWC) adjusts fusion layer weights based on modal quality scores (e.g., prioritizing textual features when visual resolution $\leq 300\text{dpi}$), reducing alignment error by 12% [20, 71]; Domain-Adaptive Pre-training (DAP) with vertical-field image-text pairs (e.g., medical CT-text datasets) enables the fusion layer to learn low-resolution feature encoding, cutting information loss to 10–15%. Both methods balance efficiency and accuracy without additional computational overhead. In clinical cross-modal tasks (e.g., low-resolution CT-text diagnosis), DAP-integrated models reduce misalignment between pathological descriptions and visual features by 30%, and DWC improves lesion recognition accuracy by 18% compared to baseline multimodal models (GPT-4: 65% $\rightarrow$ DWC+DAP: 83%). For audio-visual fusion in surveillance, DWC cuts

temporal misalignment error by 22%, meeting real-time processing requirements (latency ≤ 200 ms). Cross-modal alignment is constrained by three critical algorithmic limitations: first, fusion layer information loss (20–30% in low-resolution scenarios) arising from inherent semantic gaps between visual and textual features [71], which impairs effective feature integration; second, modal imbalance caused by insufficient vertical domain pre-training data (e.g., medical image-text pairs accounting for $\leq 3\%$ of corpora), leading to biased weight learning toward dominant modalities; third, temporal misalignment in audio-visual tasks—gaps exceeding 10 ms reduce semantic extraction accuracy by 15%—a limitation inherent to Transformer’s sequential processing mechanism that hinders real-time cross-modal synchronization.

3.4 Performance and potential of LLMs in cross modal tasks

Multimodal self-instruction techniques boost audio-visual fusion: Wang et al. align language models with self-generated multimodal instructions [71], enabling LLMs to interpret visual and audio information contextually—this improves video understanding accuracy by 22% and supports tasks like image-text matching and video commentary generation (e.g., CCTV listening model mentioned in).

In terms of audio-visual fusion, LLMs can process and analyze both image and audio information in videos to accomplish more complex and refined tasks. For instance, in video understanding tasks, LLMs can recognize objects, scenes, and actions in video frames, as well as the voice content in audio tracks, thereby grasping the overall meaning and emotional tone of the video. The CCTV listening model is a typical example: it can generate text content such as commentaries and titles based on video content, and even produce high-quality videos according to user requirements.

While GPT-4’s multimodal capability achieves unified text-visual processing (Section 2.3), it still has modal bias—the XLM-R model mentioned in Sect. 2.3, which supports over 100 languages in cross-lingual generalization, shows higher dependence on visual resolution in medical imaging tasks. Specifically, in low-resolution CT image diagnosis, the lesion recognition accuracy of Encoder-Only model BioBERT (Sect. 3.4.1) is 5% higher than that of GPT-4. Additionally, though the CCTV listening model improves video interpretation accuracy by 22%, its ‘high computational cost’ defect (Sect. 3.3) requires integration with model compression techniques (Sect. 2.5.7) for edge device deployment: 8-bit quantization can reduce computational complexity by 40%, but leads to a 3% drop in audio semantic extraction accuracy, a trade-off that must be judged based on scenario priorities (e.g., community security prioritizes

real-time performance over minor accuracy loss). This trade-off aligns with the modal alignment optimization discussed in Sect. 3.3—quantization balances efficiency and accuracy, while DAP/DWC further mitigates accuracy loss in low-resolution scenarios.

In the realm of text-to-image generation, LLMs can automatically create matching images based on input text information. For example, text-to-image technology (known as Wenshengtu in Chinese) is built on deep learning algorithms—especially GANs and diffusion models. Through training on large-scale datasets, AI can understand text descriptions and generate corresponding images. Many models at home and abroad have achieved remarkable results in this field, such as Baidu’s ERNIE knowledge-enhanced model, and international models like MidJourney and Stable Diffusion.

LLMs hold significant potential to enhance information retrieval efficiency, drive the development of the creative industry, enable more intelligent human-computer interaction, and expand application scenarios in cross-modal tasks. They can support functions such as querying images via text or retrieving relevant text from images, thereby improving the efficiency and accuracy of information retrieval. In the creative industry, LLMs can generate images or videos that meet user needs based on descriptions or ideas. In smart home and intelligent customer service fields, LLMs enable intelligent interaction by recognizing users’ voice commands and actions. With continuous technological progress and innovation, the application scenarios of LLMs in cross-modal tasks will continue to expand.

The adaptability of current LLM cross-modal technology routes (audio-visual fusion, text-to-image) varies significantly, and trade-offs must be made based on scenario requirements:

The audio-visual fusion route (e.g., CCTV listening model) has its advantage in collaborative understanding of multimodal information—it can synchronously analyze video frames (object recognition) and audio (speech content), which improves the accuracy of video commentary generation by 22%. However, it has a limitation of high computational cost (needing to process both image and audio features), which makes it only suitable for high-performance server scenarios (e.g., media content production). If applied to edge monitoring devices (e.g., community security), a lightweight solution of key frames with keyword extraction is recommended, where partial accuracy is sacrificed for real-time performance.

The text-to-image route (e.g., Stable Diffusion, Baidu ERNIE-ViLG) excels in generation diversity—it supports generating stylized images (e.g., illustrations, product design drafts) based on text descriptions. Yet, it faces the risk of semantic deviation (e.g., an approximately 5% probability of generating a green apple when the text specifies

a red apple), which makes it more appropriate for creative design scenarios (e.g., advertising material production). If used in precision-matching scenarios (e.g., e-commerce product image generation), text-image semantic alignment verification (e.g., similarity calculation based on the CLIP model) should be integrated to ensure a matching rate of $\geq 90\%$.

The cross-modal retrieval route (e.g., text-to-image search, image-to-text search) stands out for information retrieval efficiency—it can break through the limitations of single-modal retrieval (e.g., searching for unlabeled images using text). Nevertheless, it relies on large-scale cross-modal datasets, and its performance drops sharply (with accuracy decreasing from 85% to 60%) in niche fields (e.g., ancient book image retrieval). To improve its adaptability, domain fine-tuning (e.g., integrating ancient text-image paired data) is required.

Although LLMs demonstrate great potential in cross-modal tasks, they still confront challenges including diverse data modalities, multimodal data mismatch, modal fusion and alignment, computational resource constraints, and training complexities. In order to solve these challenges, we should strengthen the construction and sharing of cross modal data sets, research and develop more efficient algorithms and technologies, explore distributed computing and other technical means to reduce the computational cost, and strengthen the research on the interpretability and robustness of the model.

3.5 Other applications

3.5.1 Healthcare sector

To clarify conceptual boundaries and avoid task-modal confusion in subsequent discussions, we first define three core model categories for medical imaging analysis: pure LLMs take text-only data (e.g., electronic health records, radiology reports, and biomedical literature) as their exclusive input, with core capabilities covering medical language

understanding, clinical text classification, and named entity recognition (NER) of medical terminologies, and their application in medical imaging is restricted to indirect auxiliary tasks (e.g., matching lesion descriptions in reports with standard terminology, summarizing radiology findings) without direct image processing; multimodal LLMs support dual or multi-source input of text and image, featuring core capabilities including cross-modal feature fusion, image-text alignment, and visual question answering (VQA), and they are applied in medical imaging for joint analysis tasks (e.g., answering clinical questions about medical images with text prompts, generating radiology reports from image inputs); vision foundation models (VFMs) accept image-only input (e.g., X-rays, CT scans, MRIs), focus on core capabilities such as visual feature extraction, lesion detection, pixel-level segmentation, and pure image diagnosis, and are dedicated to end-to-end visual diagnostic tasks in medical imaging (e.g., lung nodule detection in chest CT scans, tumor segmentation in MRI images). The mapping relationships between specific medical imaging tasks, data modalities and applicable models are systematically summarized in Table 4.

The core value of LLMs in healthcare resides in multimodal data integration, analysis, and improved efficiency of medical workflows. By fusing multi-source data such as electronic health record (EHR) texts, medical images (e.g., CT, MRI), and gene sequences, LLMs support the entire workflow from auxiliary diagnosis to drug development. For instance, in clinical diagnosis scenarios, LLMs can integrate patients' medical history texts with imaging reports to generate structured diagnostic recommendations; in drug development, they accelerate target screening and clinical trial data analysis, shortening the research and development cycle [72].

However, Multimodal LLMs still face prominent limitations in pure visual diagnostic tasks (e.g., independent lung nodule segmentation from CT scans), where vision foundation models maintain inherent advantages [73], and found that their accuracy on such professional visual tasks only ranged from 44.6% to 55.4%, far lower than their performance on text-based medical questions (68.3%–79.1%) [73]. GPT-4o, as a multimodal LLM, achieves 20% accuracy in orthodontic instrument recognition with text prompts, while the vision foundation model ResNet-50 achieves 89% accuracy in the pure visual version of the same task [66]. In contrast, BioBERT achieves 65% accuracy in the same task [73], attributed to its domain-adapted pre-training on 1.2M biomedical text-image pairs—this 45-percentage-point gap underscores the critical role of specialized data in overcoming architectural limitations of multimodal models. Integrating DAP (Section 3.3) into multimodal models narrows this gap to 12%, verifying the effectiveness of domain-adaptive

Table 4 Task-Modality-Model Mapping in Medical Imaging

Task Type	Modality	Model	Metric	Example
Report Standardization	Text	Pure LLMs	Terminology Acc	Align ICD-10 codes
Image-text Diagnosis	Text+Image	Multimodal LLMs	VQA Acc	CT nodule QA
Pure Visual Lesion Detection	Image	VFMs	Sensitivity	Lung nodule detection
Report Generation	Text+Image	Multimodal LLMs	Consistency	X-ray report generation

optimization. This reveals the architectural bottleneck of general multimodal LLMs in parsing domain-specific medical visual features. This bottleneck leads to critical failure modes: 27.4% hallucination in drug interaction predictions (60% posing clinical risks) [74]. A dual safeguard—domain knowledge base integration and professional physician review—is essential to mitigate harms. Additionally, based on personalized health data (e.g., living habits, vital sign monitoring), LLMs can generate customized health management plans to facilitate disease prevention.

3.5.2 Fintech sector

LLMs provide fintech with core capabilities for intelligent decision-making and risk management, covering three key scenarios: investment, customer service, and risk control. In investment decision-making, LLMs can analyze multi-dimensional market text data (e.g., financial reports, news, research papers) to generate quantitative analysis reports; in intelligent customer service, they support 24/7 multilingual conversational interactions to address user needs such as account inquiries and business processing; in risk control, they identify abnormal transaction patterns by mining transaction records and user behavior data, reducing fraud risks [75]. However, LLMs in fintech scenarios face non-negligible algorithmic bias risks: credit scoring models trained on historical economic data may reject loan applications from specific low-income regions at a rate 28% higher than the average [76], stemming from the encoding of historical regional economic disparities in training corpora rather than actual creditworthiness of applicants. To mitigate such biases and balance risk control with financial inclusion, it is necessary to introduce fairness regularization to constrain model weight distribution and build a dynamic bias monitoring system for real-time calibration of credit approval results, ensuring that the model's decision-making logic aligns with both business risk requirements and social equity principles. Practical mitigation techniques include adversarial debiasing (AD), which trains the model to resist region-related bias features, reducing regional loan rejection disparities by 18% without compromising predictive accuracy [77]; and Counterfactual Data Augmentation (CDA), which generates synthetic samples for underrepresented regions to balance training data distribution, improving demographic parity by 12% in credit scoring models [78]. A long-term monitoring mechanism—including monthly bias audits (tracking demographic parity $\leq 5\%$) and real-time adjustment of feature weights—can maintain regional bias below 5% for over 18 months, compared to 6-month drift in models without such safeguards. For example, a major fintech platform adopted this mechanism and reduced regional loan rejection disparities from 28% to 7% within 3 months.

3.5.3 Education sector

The core value of LLMs in reshaping educational scenarios lies in personalized teaching and automated assessment. Based on students' learning data (e.g., homework completion, mastery of knowledge points), LLMs generate customized learning paths—for example, providing targeted grammar correction and writing guidance for language learners. In the assessment process, LLMs can automatically grade homework, conduct paper plagiarism checks, and generate multi-dimensional feedback (e.g., logical completeness, knowledge gaps) [79]. Notably, in early childhood vocabulary learning, multimodal LLMs show significant potential but also non-negligible limitations: Hang et al. [80] verified based on dual coding theory that personalized image-text flashcards generated by multimodal LLMs can improve young children's vocabulary memory rate by 32% compared with traditional prefabricated flashcards; Notably, visual complexity exceeding children's cognitive threshold reduces comprehension accuracy by 18%, highlighting the need for simplified visual elements (≤ 2 core features) in educational multimodal content. Age-adaptive design further enhances effectiveness: for 3-5-year-olds, visual elements should use high-contrast colors and single simple shapes; for 6-8-year-olds, abstract concepts can be introduced but limited to 1-2 per flashcard—this refinement improves vocabulary comprehension by an additional 12% compared to one-size-fits-all multimodal content. This conclusion provides empirical evidence for the content adaptation of multimodal LLMs in early childhood education and highlights the core value of “visual-language balance” in low-age education scenarios. Furthermore, LLMs can serve as virtual tutors, answering students' questions through interactive conversations and breaking the constraints of time and space.

3.6 Machine learning

3.6.1 Few-shot learning and transfer learning

To alleviate data scarcity in LLMs, FSL and transfer learning are widely adopted, as they reduce reliance on large-scale annotated datasets. FSL addresses this issue by enabling models to learn effective task-solving patterns with a small number of labeled samples. In order to alleviate this problem, technologies such as FSL and transfer learning are widely used in LLMs.

FSL addresses this issue by enabling models to learn effective task-solving patterns with a small number of labeled samples. It leverages high-dimensional feature representation capabilities and meta-learning strategies to extract generalizable knowledge from limited data, reducing

the reliance on massive annotated datasets. For example, in medical text diagnosis tasks, FSL can enable LLMs to recognize disease-related text features and achieve preliminary diagnostic results with only dozens of annotated medical record samples. FSL collaborates with transfer learning to alleviate data scarcity: for instance, Koala uses transfer learning to inherit pre-trained dialogue knowledge, then fine-tuning with more than 100 annotated dialogue samples (FSL), cutting annotation demand by 60% while maintaining dialogue coherence.

Transfer learning not only reduces the demand for annotated data in the target domain but also shortens the model training cycle and improves training efficiency. In the field of legal text analysis, for example, a pre-trained LLMs on a general text corpus can be fine-tuned with a small number of labeled legal documents through transfer learning, quickly adapting to legal text classification, named entity recognition, and other tasks, without the need to train a model from scratch using a large number of legal annotated datasets.

The combination of FSL and transfer learning further expands the application scope of LLMs. By integrating the advantages of both-transfer learning provides a solid knowledge foundation, and FSL optimizes task adaptation with limited data-LLMs can better adapt to scenarios where annotated data is scarce, such as small and medium-sized enterprise customer service dialogue systems, regional dialect text processing, and other fields, promoting the popularization and application of LLMs technology in more specialized and niche domains.

3.6.2 Model compression and lightweighting

Deploying LLMs in resource-constrained environments necessitates model compression techniques, with three predominant approaches: knowledge distillation [67], quantization [68], and pruning [81]. Quantization achieves the highest compression rate (75-87%, Table 5) but degrades FSL accuracy by 5-8% (Section 3.5.1)-a limitation not seen in knowledge distillation [67, 68], which preserves 97% performance but only reduces model size by 40%. Pruning [81], meanwhile, balances compression (20-40%) and accuracy but requires complex sensitivity analysis to avoid removing critical parameters. For edge devices (e.g., mobile), the lightweight method is preferred despite accuracy loss [68]; for cloud deployment (e.g., medical diagnosis), the high-precision method is more reliable due to its high accuracy retention [67]. A detailed comparative analysis of these three model compression techniques is presented in Table 5.

As shown in Table 5, quantization technology achieves the highest compression rate (75-87%). However, in FSL scenarios, its accuracy decreases by 5-8% (see Section 3.5.1). Therefore, the selection of compression technology

Table 5 Comparative Analysis of Model Compression Techniques

Technique	Mechanism	Com- pression Rate	Speedup	Best Application Scenarios
Knowledge Distillation	Transfer teacher knowledge to student	30-50% [67, 82]	1.5x-2.0x [67, 82]	Cloud deployment requiring accuracy
Quantization	Reduce numerical precision	75-87% [68, 83]	2.0x-3.0x [68, 83]	Edge devices with memory limits
Pruning	Remove redundant parameters	20-40% [81, 84]	1.5x-2.5x [81, 84]	Accuracy-critical applications

Table 6 Analysis of RAG and Agent Paradigms

Aspect	RAG Systems	Agent Systems
Core Innovation	Knowledge grounding	Autonomous execution
Primary Strengths	Factual accuracy	Planning and tool use
Key Challenges	Retrieval quality	Error propagation
Architectural Limits	Context windows	Reasoning depth
Current Solutions	Adaptive retrieval	Hierarchical planning

should be based on resource constraints and task accuracy requirements-quantization is preferred for edge devices, while knowledge distillation is prioritized for cloud deployment.

The selection among these techniques involves critical trade-offs that practitioners must consider:

Knowledge Distillation achieves the best balance between compression and performance retention, making it ideal for cloud-based applications where inference accuracy cannot be compromised. However, it requires access to a high-quality teacher model and substantial computational resources for the distillation process. The technique is particularly effective for maintaining semantic understanding and contextual awareness in natural language processing tasks.

Quantization provides the most significant model size reduction, crucial for mobile and embedded systems with strict storage constraints. The primary limitation is potential accuracy degradation, especially at lower bit-widths (e.g., 4-bit quantization), which may be unacceptable for certain precision-sensitive tasks. This approach excels in scenarios where memory footprint and power consumption are primary concerns, such as mobile applications and IoT devices.

Pruning offers moderate compression while best preserving the original model's capabilities, particularly effective for reducing computational FLOPs. The challenge lies in identifying truly redundant parameters without affecting critical model functionality, requiring careful sensitivity analysis. Pruning is well-suited for applications where

computational efficiency is paramount but complete accuracy preservation is necessary.

In practice, these techniques are often combined—for instance, applying pruning followed by quantization—to achieve compounded benefits. The optimal strategy depends on specific deployment constraints: quantization for strict size limits, distillation for accuracy preservation, and pruning for computational efficiency. Hybrid approaches can achieve compression rates exceeding 90% while maintaining over 95% of original model performance for many practical applications. Among single compression techniques, pruning is less suitable for cross-modal fusion layers—removing 10% of fusion layer parameters reduces alignment accuracy by 8% due to disrupted modal semantic coupling. For text-image generation tasks, 8-bit quantization (87% compression rate) maintains 92% semantic consistency (CLIP similarity score ≥ 0.85), making it suitable for edge-device deployment (e.g., mobile design tools); in audio-visual fusion (CCTV listening model), knowledge distillation preserves 95% audio semantic accuracy and cuts inference latency by 60%, ideal for cloud-based real-time processing. These findings are consistent with empirical results from knowledge distillation and AWQ quantization [82, 85], which verify that compression technique selection must be task-specific to avoid alignment degradation in cross-modal fusion layers. This aligns with multimodal compression best practices—prioritizing quantization or distillation over pruning for fusion-sensitive tasks to balance efficiency and alignment quality.

The mathematical foundations of these techniques further illustrate their respective strengths and limitations. Knowledge distillation employs a composite loss function that balances task accuracy with feature alignment [82]:

$$\mathcal{L}_{KD} = \alpha \mathcal{L}_{CE} + (1 - \alpha) \mathcal{L}_{MSE} \quad (10)$$

where: $\mathcal{L}_{KD}$: Total knowledge distillation loss that integrates two complementary supervision signals for student model training. This combined loss ensures the student learns both task-specific accuracy and the teacher's rich feature representations. $\mathcal{L}_{CE} = -\sum y \log(\hat{y}_{\text{student}})$: Cross-entropy loss that calculates the discrepancy between the student model's predictions ($\hat{y}_{\text{student}}$) and the ground-truth labels (y). This term ensures the student model maintains basic task-solving capabilities and classification accuracy. $\mathcal{L}_{MSE} = \text{MSE}(h_{\text{student}}, h_{\text{teacher}})$: Mean squared error loss that aligns the intermediate hidden states of the student (h_{student}) and teacher (h_{teacher}). This enables the student to learn the teacher's knowledge-subtle feature representations and internal patterns that are not captured by the final output alone. $\alpha = 0.5$: Balance hyperparameter that adjusts the relative weight of the two loss components. When α is

higher, the training prioritizes task accuracy; when lower, it emphasizes knowledge transfer from the teacher's internal representations.

For language models, temperature scaling (τ) is applied to soften the teacher's output distribution, which preserves richer semantic information compared to hard labels. The softened probability distribution is computed as [82]:

$$p_{\text{teacher}} = \frac{\exp(z_{\text{teacher}}/\tau)}{\sum \exp(z_{\text{teacher}}/\tau)} \quad (11)$$

where: p_{teacher} : Softened probability distribution of the teacher model that contains richer semantic information and fine-grained class relationships. z_{teacher} : Logits (raw output values before applying softmax) of the teacher model, representing the model's confidence scores for each class. τ : Temperature parameter that controls the smoothness of the output distribution. Higher τ values produce softer distributions that better preserve relative class relationships. $\exp(\cdot)$: Exponential function that converts logits to positive values suitable for probability calculation.

Quantization relies on precision reduction through scaling and rounding operations [83]:

$$\text{scale} = \frac{\max(|\text{weight}|)}{127}, \quad w_{\text{int8}} = \text{round}(w_{\text{fp32}}/\text{scale}) \quad (12)$$

where: scale : Quantization scaling factor derived from the maximum absolute value of original 32-bit floating-point weights. This ensures the full INT8 range $[-127, 127]$ is utilized without clipping extreme values. w_{fp32} : Original 32-bit floating-point weights of the model, providing high precision but requiring significant storage. w_{int8} : Final 8-bit integer weights after scaling and rounding, reducing storage requirements by 75% while maintaining compatibility with most hardware. $\max(|\text{weight}|)$: Maximum absolute value in the weight tensor, used to determine the appropriate scaling factor for the quantization process. $\text{round}(\cdot)$: Rounding function that converts continuous floating-point values to discrete integers while minimizing quantization error.

Advanced quantization techniques like AWQ (Activation-Aware Weight Quantization) minimize quantization loss by prioritizing weights that significantly impact activation outputs [85]:

$$\mathcal{L}_{\text{AWQ}} = \sum |\text{Activation}(W_{\text{quantized}}) - \text{Activation}(W_{\text{fp16}})|^2 \quad (13)$$

where: $\mathcal{L}_{\text{AWQ}}$: AWQ quantization loss that measures the difference between activations produced by quantized and original weights. $\text{Activation}(W_{\text{quantized}})$: Model's activation outputs computed using quantized weights, representing the actual behavior after compression. $\text{Activation}(W_{\text{fp16}})$:

Model's activation outputs computed using original 16-bit floating-point weights, serving as the reference for maintaining model functionality. $|\cdot|^2$: Squared difference that penalizes larger discrepancies more heavily, ensuring the quantized model closely matches the original's behavior.

Pruning techniques employ importance metrics to identify redundant components. Gradient-based pruning identifies critical attention heads using [81]:

$$\text{Importance}(h_i) = \mathbb{E} \left| \frac{\partial \mathcal{L}}{\partial h_i} \right| \quad (14)$$

where: $\text{Importance}(h_i)$: Importance score of the i -th attention head, quantifying its contribution to task performance. $\frac{\partial \mathcal{L}}{\partial h_i}$: Partial derivative of the task loss with respect to the attention head's outputs, indicating how much the loss changes with small changes to the head's activity. $\mathbb{E}[\cdot]$: Expectation over training data, ensuring the importance score reflects average contribution across diverse inputs. $|\cdot|$: Absolute value that considers both positive and negative gradients as indicative of importance.

Structured pruning methods like LayerDrop randomly drop layers during training [84]:

$$\mathcal{L}_{\text{LayerDrop}} = \mathbb{E}_{m \sim \text{Bernoulli}(p)} \mathcal{L}(f_m(W, x)) \quad (15)$$

where: $\mathcal{L}_{\text{LayerDrop}}$: LayerDrop training loss that accounts for random layer removal, encouraging robustness to architectural changes. $m \sim \text{Bernoulli}(p)$: Random variable following Bernoulli distribution where each layer has probability p of being dropped during training. $f_m(W, x)$: Model function using m retained layers, with W representing weights and x representing input data. $\mathbb{E}[\cdot]$: Expectation operator that averages the loss over different layer-dropping configurations, ensuring the model learns to perform well under various architectural conditions.

Empirical studies demonstrate these trade-offs in real-world scenarios. For example, in mobile deployment, 8-bit quantization reduces LLaMA-7B model size by 75% while maintaining 98% of original performance on language understanding tasks. In medical diagnosis applications, knowledge distillation proves more suitable, with distilled models retaining 97% of diagnostic accuracy while achieving 60% faster inference compared to the original BERT-base model. For real-time autonomous systems, pruned models demonstrate 40% reduction in computational requirements while maintaining critical safety margins.

Despite their benefits, all compression techniques face inherent limitations. Quantization struggles with outlier activations that can disproportionately affect model output, particularly in FSL scenarios. Pruning may remove parameters that appear redundant during training but become

important for certain inference inputs or edge cases. Knowledge distillation is constrained by the quality and diversity of the teacher model's knowledge, and its effectiveness diminishes when the student model architecture differs significantly from the teacher.

These limitations highlight that compression is not a one-size-fits-all solution but requires careful evaluation against target application requirements. Practitioners must consider multiple factors including accuracy thresholds, latency constraints, hardware capabilities, and the specific nature of the target tasks. Ongoing research focuses on developing adaptive compression methods that can dynamically adjust compression levels based on real-time performance requirements and resource availability.

As LLMs continue to evolve, compression techniques must adapt to new architectural innovations and application demands. The integration of compression-aware training methodologies and the development of hardware-aware compression strategies represent promising directions for future research in this critical area of model optimization.

3.6.3 Increased interpretability

Improving model interpretability is an important research direction in the field of machine learning. It is helpful for humans to better understand the decision-making process and output results of the model, so as to optimize and improve the model. Current methods to improve the interpretability of models mainly include attention visualization and model distillation.

Attention visualization improves model interpretability by revealing which input features the model attends to during decision-making. Methods include activation maps, feature mapping, and attention matrix visualization. These visualization tools can intuitively show the attention of the model to different input characteristics in the decision-making process, and help researchers and developers better understand the decision-making process of the model.

Model distillation enhances interpretability by distilling knowledge from a complex LLM into a smaller, simpler model. By training a small model to imitate the output or middle layer characteristics of a LLMs, knowledge transfer and model simplification can be realized. This method can not only reduce the complexity and computation of the model, but also improve the interpretability and ease of use of the model.

Beyond attention visualization and model distillation, numerous other methods exist to enhance model interpretability. For example, feature importance analysis can evaluate the importance of features by calculating the contribution of each feature to the model output [86]; Visual interpretation can show the decision-making process and output

results of the model by generating visual charts or animations; The explanatory model can use a simpler and more intuitive model to replace the complex black box model; The interpretability algorithm can consider the requirements of interpretability when designing the algorithm; User interaction and feedback can improve the interpretability and user experience of the model by interacting with users and obtaining feedback [87].

3.6.4 Privacy protection and security mechanisms

Technologies such as differential privacy, federated learning and adversarial training provide strong support for data privacy protection and model security.

Differential privacy is a privacy-preserving technique with a rigorous mathematical definition. It prevents the disclosure of individual private information by adding carefully calibrated noise to query results [88]. Differential privacy introduces a concept called privacy loss or privacy budget parameter, which is used to control the amount of noise or randomness added to the original dataset. By adding noise, the attacker cannot accurately infer the individual information in the original data even if they obtain the noisy data. Differential privacy can be applied to various scenarios requiring data privacy protection, such as finance, healthcare, and artificial intelligence [89].

Federated learning is a distributed machine learning paradigm [90]. It provides a solution for data privacy preservation. In federated learning, multiple participants collaborate to train machine learning models without sharing raw data. Each participant retains their data and conducts local model training. These local model parameters are sent to a central server for aggregation to generate a global model. Through multiple iterations, a high-quality global model can be obtained without exposing participants' private data. Federated learning is widely used in privacy-sensitive scenarios like finance, healthcare, and IoT [91].

Adversarial training is a method to enhance model robustness and security [69]. It improves the model's robustness against adversarial attacks by training with adversarial samples. Adversarial attacks involve attackers using carefully designed input samples (adversarial examples) to deceive models into making incorrect predictions or classifications. Adversarial training enables models to learn to resist these attacks by incorporating adversarial samples into the training process. Techniques include adversarial sample generation, model training, and iterative optimization. Adversarial training is widely used in deep learning models for image recognition, speech recognition, natural language processing, etc., enhancing model security by improving resistance to adversarial attacks.

For offline RLHF-where pre-collected preference data may be corrupted by malicious actors (e.g., feedback flipping or trajectory tampering)-a corruption-robust framework (AISTATS 2025) provides provable security guarantees: it learns a reward model with confidence sets from contaminated data and optimizes a pessimistic policy within these sets, ensuring that even when 15% of preference data is corrupted, the model retains 92% of its performance in medical diagnosis RLHF tasks [92].

By adopting effective coping strategies and cutting-edge technologies, we can overcome data scarcity and high labeling costs, promoting continuous progress and innovation in AI. Strengthening privacy protection and security mechanisms is also critical to ensuring AI's healthy development. With technological advancements, AI will play a greater role in more fields, contributing to human progress [88].

3.7 Emerging paradigms: retrieval-augmented generation and AI agent systems

The rapid evolution of LLMs has catalyzed the development of two transformative paradigms: retrieval-augmented generation (RAG) and AI agent systems. These architectures extend LLMs beyond their original text generation capabilities to more robust, knowledge-grounded, and interactive systems [1, 93].

3.7.1 Retrieval-augmented generation (RAG)

Retrieval-augmented generation (RAG) addresses parametric knowledge limitations via a three-stage architecture, starting with dense passage retrieval using dual-encoder models [94]. While this dual-encoder model design improves open-domain QA accuracy by 30% (Section 3.6.1), it struggles with retrieval noise—expanding the document scope increases irrelevant content by 20%, reducing answer accuracy by 15% [94]. This limitation is partially mitigated by hybrid retrieval, which combines the aforementioned dense matching with sparse keyword search to cut noise by 8%, though this hybrid retrieval method adds 10% inference latency [68]. Thus, the dual-encoder model is suitable for small-scale knowledge bases (e.g., enterprise FAQs) while the hybrid retrieval method is better for large-scale corpora (e.g., academic paper retrieval) [68, 94].

Despite their promising capabilities, RAG systems face substantial technical challenges. The retrieval-quality trade-off presents a fundamental constraint, where expanded document retrieval introduces noise and computational overhead while restricted search strategies risk missing essential information [95]. Compositional reasoning deficiencies persist as models frequently fail to synthesize information across multiple documents or identify subtle contradictions

in retrieved evidence [96]. The temporal knowledge lag further constrains practical effectiveness, as retrieval corpora require periodic updates, creating intervals of outdated information [97]. Most critically, RAG does not adequately address the reasoning verification problem, as models may generate plausible but incorrect conclusions even when provided with contradictory evidence [98].

3.7.2 AI agent systems

Parallel to these developments, AI agent systems transform LLMs from passive text generators into autonomous problem-solving entities. Modern agent frameworks integrate reasoning engines that decompose complex goals into executable sub-tasks [99], hierarchical memory systems maintaining both short-term context and long-term experience [100], tool-use interfaces enabling API calls and environment interaction [101], and reflection mechanisms that evaluate and verify intermediate outcomes [102]. These integrated systems demonstrate emergent capabilities in multi-step planning, tool composition, and adaptive strategy formulation.

However, agent systems confront distinct reliability and safety concerns. Error propagation in multi-step tasks presents a fundamental challenge, where initial mistakes compound through subsequent actions with amplifying effects [103]. Compositional generalization limits mean that agents trained on individual tools often fail when novel tool combinations are required [104]. The safety-trust dilemma emerges prominently as agents gain broader environmental access, where comprehensive safety verification proves computationally prohibitive while restricted tool sets limit functional utility [105]. Resource management inefficiencies further constrain practical deployment, with agents frequently engaging in computationally expensive trial-and-error behaviors [106]. The error propagation issue of AI Agent can be alleviated by the RLHF reflection mechanism (Sect. 2.5.4): in multi-step tasks, after each sub-task is completed, the Reward Model is called to verify results-if

the error rate exceeds 10%, backtracking adjustment is performed, reducing the final task error rate from 35% to 12%, though increasing inference time by 25%. For insufficient tool combination capability, transfer learning (Sect. 3.5.1) can be integrated to share 30% of parameters between single-tool training knowledge (e.g., text translation, image generation), which improves the success rate of cross-tool tasks by 18% (referencing the Koala model's transfer learning case in Sect. 3.5.1). The core capabilities and corresponding challenges of the AI Agent paradigm across key dimensions are summarized in Table 7.

Practical implementation of AI Agent systems relies on synergy with reinforcement learning and transfer learning. First, the RLHF reflection mechanism mitigates error propagation: in multi-step tasks, after each subtask completion, the Reward Model (e.g., DeepSeek's dual-tower model) verifies results—if the error rate exceeds 10%, backtracking adjustment is triggered, reducing the final task error rate from 35% to 12% (though inference time increases by 25%). Second, transfer learning enhances tool combination capabilities: by sharing 30% of parameters from single-tool training (e.g., text translation, image generation), the success rate of cross-tool tasks (e.g., “translate text and generate corresponding image”) improves by 18%—consistent with the Koala model's transfer learning case of reducing annotation demand via knowledge inheritance. In cross-modal scenarios (e.g., ‘analyze medical images and generate diagnosis reports’), AI Agent integrates DWC (Sect. 3.3) and transfer learning to complete tool chaining: it first uses BioBERT for image feature extraction, then invokes medical NLP tools for report generation, reducing manual intervention by 50% and improving report consistency by 25% (F1 score ≥ 0.91) compared to standalone multimodal models.

Architecturally, both paradigms expose core constraints of contemporary LLMs designs. RAG systems struggle with inherent context window limitations, as the combination of retrieved documents and generation context frequently exceeds available model capacity [107]. Agent systems reveal reasoning depth boundaries, with performance degrading sharply as planning horizons extend beyond several steps [108]. The integration overhead of maintaining these auxiliary systems often negates the efficiency gains achieved through model compression techniques [109].

Current research directions seek to address these limitations through adaptive retrieval mechanisms [110], verification-enhanced generation frameworks [111], hierarchical planning approaches [112], and safe exploration policies [113]. The integration of formal verification methods with neural generation represents another frontier for addressing fundamental trustworthiness gaps [114]. However, these approaches must overcome substantial scalability and generalization challenges before achieving robust practical

Table 7 Analysis of the AI Agent Paradigm

Aspect	Capabilities	Challenges
Task Automation	Automates complex workflows and processes	Error propagation in multi-step task sequences
Generalization	Adapts tools for novel task requirements	Limited performance with unfamiliar tool combinations
Autonomy	Independent operation with minimal oversight	Safety risks in real-world environment interaction
Resource Management	Strategic tool usage to enhance capabilities	Inefficient trial-and-error search behaviors

implementation, indicating that current solutions primarily address symptoms rather than underlying architectural deficiencies in LLMs Table 7.

Based on the decision framework in Table 8, the model selection process follows three core steps: first, define the core task type by prioritizing task attributes such as visual-only versus cross-modal and generation versus understanding to narrow down architectural candidates, for instance, selecting vision foundation models (VFM) for pure visual tasks and Encoder-Only architectures for named entity recognition (NER); second, balance resource constraints, where limited computational resources (e.g., for edge devices) call for lightweight solutions like quantization or pruning paired with Encoder-Only or compressed multimodal models, while cloud deployment with high accuracy demands favors full-parameter Decoder-Only models or mixture-of-experts (MoE) architectures; third, optimize the training strategy, as safety-critical scenarios (e.g., healthcare) benefit from domain-adaptive pre-training (DAP) combined with reinforcement learning from human feedback (RLHF) with a small γ parameter to reduce hallucination, and rapid deployment scenarios (e.g., legal text analysis) leverage supervised fine-tuning (SFT) alongside transfer learning to cut annotation costs. This decision logic aligns with the "performance-efficiency-cost" trade-off core of this survey, ensuring that model selection is not only technically feasible but also practically applicable to specific scenario constraints.

4 Challenges

4.1 The cost of data annotation and its countermeasures in the development of LLMs

The high annotation cost is primarily reflected in the significant time and human resources required for the data annotation process. In the machine learning pipeline, annotating data is a core step for training models, but this process is often extremely labor-intensive. Take the sentiment analysis task in the field of natural language processing (NLP) as an example. Manual annotation not only requires significant time to judge the emotional tendency of texts sentence by sentence but also suffers from subjectivity in semantic understanding among different annotators, leading to inconsistent annotation quality [115]. These issues directly impact model training performance and may even cause the model to provide erroneous emotional judgments in practical applications. Additionally, in highly specialized fields such as bioinformatics and legal document processing, annotation work requires annotators to possess profound professional knowledge and skills, further increasing the annotation costs.

Table 8 Scenario-Constraint-Model Decision Framework for LLMs

Application Scenario	Core Task Type	Resource Constraints	Recommended Architecture	Training Strategy	Deployment Solution
Healthcare (Medical Imaging)	Visual diagnosis + Report generation	High accuracy; Low hallucination	VFM (pure visual) + Multimodal LLMs (cross-modal)	DAP + RLHF ($\gamma=0.15$)	Knowledge distillation (cloud)
Fintech (Credit Scoring)	Risk prediction + Fairness requirement	Low bias; Real-time inference	Encoder-Only (BERT)	Fairness regularization + SFT	Quantization (8-bit, edge)
Education (Early Childhood Learning)	Multimodal content generation	Low complexity; High interpretability	Multimodal LLMs (lightweight)	Instruction fine-tuning	Pruning (40% compression)
Autonomous Driving	Object detection + Scene analysis	Real-time; Low latency	Encoder-Decoder + VFM	Transfer learning + FSL	Edge-cloud collaboration
Enterprise FAQs	Open-domain QA	Low cost; Low hallucination	Decoder-Only + RAG	SFT	Quantization (4-bit)
Creative Content Generation	Open-ended text/image generation	High diversity; Flexible output	Decoder-Only (GPT series)	RLHF ($\gamma=0.2$) + Prompt learning	Cloud deployment (full parameter)
Legal Text Analysis	Text classification + NER	Fast adaptation; Low annotation cost	Encoder-Only (RoBERTa)	Transfer learning + FSL	Knowledge distillation
Cross-modal Retrieval (Ancient Books)	Image-text matching	Low-resource; High alignment	Multimodal LLMs + RAG	Domain fine-tuning	Hybrid cloud-edge

To address the above challenges, effective strategies can be adopted across multiple dimensions. First, enhancing data collection and processing capabilities is key to breaking the deadlock [89]. On the one hand, advanced web crawler technologies can be used to accurately capture relevant data from massive amounts of information on the Internet following compliance procedures; on the other hand, IoT sensors and other devices can be utilized to collect real-time dynamic data in fields such as the environment and industrial production. Meanwhile, data cleaning can be employed to remove duplicate and erroneous information, and normalization techniques can be used to unify data formats,

significantly improving data quality and usability. Second, data augmentation techniques provide new ideas for solving the problem of data scarcity [116]. In the field of image processing, operations such as rotating, scaling, flipping, and adding noise to original images can generate a large number of new training samples, effectively expanding the scale of the dataset. In text processing, methods such as synonym replacement, sentence restructuring, and back-translation can not only increase the diversity of text data but also help models learn more semantic expression forms, enhancing the models' generalization capabilities.

Transfer learning and pre-trained models further reduce annotation costs by applying general language knowledge to specific tasks, such as adapting to legal text analysis with limited labeled data [88]. Promoting data sharing and collaboration is also an important direction in addressing data scarcity and annotation costs [90]. Consider the case of medical data-sharing platforms, which integrate case data from multiple hospitals while ensuring patient privacy, thereby providing rich data support for medical AI research and strongly promoting the innovation and application of AI technologies in the healthcare field.

For RLHF-specific annotation bottlenecks-where human preference data is costly to collect-active multi-task learning offers an efficient solution: by formulating RLHF as a contextual dueling bandit problem and allocating annotation samples based on task relevance (e.g., prioritizing high-relevance financial text tasks for fintech RLHF), this method reduces the demand for human-labeled data by 35% compared to uniform sampling while maintaining 96% of the original alignment quality [117].

Semi-supervised and unsupervised learning methods offer complementary approaches for reducing annotation costs [118, 119]. Another efficiency-focused framework improves feedback utilization by constructing sequential trajectory rankings: it iteratively selects a defender (most preferred trajectory) and a challenger (unselected trajectory) to generate preference signals, producing 2.3x more valid feedback per human annotation than traditional pairwise comparison. This cuts the annotation cost of customer service chatbot RLHF by 42% in practical deployments [120]. Semi-supervised learning combines a small amount of annotated data with a large amount of unannotated data for training, leveraging the information contained in the unannotated data to assist the annotated data in training models and effectively improving model performance. Unsupervised learning, by mining the inherent structures and patterns in data, can automatically discover latent features and patterns, providing valuable information for subsequent supervised learning tasks and reducing reliance on manually annotated data.

4.2 Hallucination phenomenon: etiology and empirical evidence of factual distortion

Factual hallucination—defined as the generation of semantically coherent but factually ungrounded content—poses a fundamental limitation to deploying LLMs in knowledge-intensive domains. Empirical evidence indicates that this phenomenon stems from dual deficiencies: the inherent noise in training data and inadequate knowledge boundaries within parametric memory. The underlying mechanisms originate from shortcomings in both training paradigms and knowledge representation. Firstly, inherent noise and annotation biases in training datasets propagate through the learning process: models prioritize linguistic fluency optimization, inadvertently amplifying these distortions. As documented, internet text contains approximately 8% erroneous information, which, when incorporated into training corpora, corrupts the acquisition of veridical knowledge [121]. Secondly, ambiguous knowledge boundaries and delayed updates characterize LLMs; when encountering domains or time-sensitive queries not covered in training data, models tend to fabricate content via statistical pattern completion rather than explicitly signaling knowledge deficits.

Empirical investigations have quantified this challenge rigorously. The work report systematically evaluated GPT-4 across academic question-answering and fact-verification tasks [65], revealing a hallucination rate of 15.8% in academic contexts-equating to nearly one in six outputs containing factual errors. Notably, in long-form text generation, this rate escalates to 32.1% when output exceeds a critical length threshold [122], reflecting the autoregressive decoding mechanism's inadequacy in maintaining global consistency, leading to cumulative error propagation [123]. Such inaccuracies are particularly consequential in high-stakes domains: medical LLMs, for instance, exhibit hallucination rates of 27.4% in drug interaction predictions, with over 60% of these errors posing potential clinical risks [74]. Domain-specific variations are notable: hallucination rates in financial report generation reach 18.2% (mostly incorrect market data), in educational content creation (textbook summaries) drop to 9.7%, and in legal document drafting climb to 23.5% (spurious case citations). This discrepancy reflects the impact of knowledge density and factual rigidity on model reliability—fields with strict factual constraints require additional retrieval-augmentation (RAG, Sect. 3.7.1) to cut hallucination by 40%–60% [74].

While RLHF can mitigate hallucination in general tasks, it often leads to reduced output diversity—a critical limitation for creative writing or academic paper generation. A curiosity-driven RLHF (CD-RLHF) framework resolves this trade-off by combining intrinsic curiosity rewards (for novel semantic states) with extrinsic human feedback: in

academic abstract generation tasks, CD-RLHF reduces hallucination rates by 8% compared to standard RLHF while increasing output diversity (measured by self-BLEU) by 15% [122]. Combining CD-RLHF with RAG (Sect. 3.7.1) achieves a synergistic effect—hallucination rates drop to 5.2% in academic writing tasks, meeting publication-level factual requirements [122].

Mitigation strategies for hallucination, including RAG and CD-RLHF, are integrated into practical scenarios like academic writing (Sect. 3.1) and medical diagnosis (Section 3.5.1).

4.3 Algorithmic bias: transmission chain from data imbalance to application risks

Algorithmic bias—manifested as systematic skews in model outputs toward specific groups or perspectives—represents a technical encoding of societal inequalities embedded in training data. This bias propagates through a cascading data-model-application pipeline rather than occurring as random noise. At the data layer, the underrepresentation of minority groups in training corpora is endemic: in law enforcement facial recognition systems, for instance, specific ethnic groups exhibit a 34% misidentification rate—over ten times higher than other demographics. This disparity arises from insufficient or biased sampling during data collection, impairing the model's ability to learn distinctive features, thereby inducing recognition inaccuracies [124].

At the model layer, statistical pattern-based learning objectives reinforce preexisting data stereotypes. Training on such imbalanced datasets entrenches biases into decision frameworks: fintech credit scoring models, this is exemplified by models that demonstrate implicit postal code sensitivity, elevating loan rejection rates for specific regions by 28%. This likely stems from historical economic disparities encoded in training data, leading to spurious associations between geographic identifiers and creditworthiness [76].

Current mitigation strategies face inherent limitations: fairness-constrained training reduces targeted biases via regularization but risks over-correction, compromising semantic coherence and general performance. Dataset reweighting techniques attempt to amplify minority sample influence but are constrained by scarcity of high-quality annotated data for underrepresented groups. These tensions underscore that bias mitigation requires dynamic equilibrium between fairness and utility rather than superficial technical fixes [77]. This equilibrium can be achieved via adaptive fairness-utility weighting: in high-stakes fintech scenarios (e.g., loan approval), fairness weights are increased to reduce regional bias below 5%; in low-risk scenarios (e.g., financial news classification), utility weights are prioritized to maintain 98% semantic comprehension accuracy.

4.4 Scalability dilemma: the scale-efficiency trade-off

The exponential growth in model scale, while driving performance improvements, has precipitated a scale-cost vicious cycle. The study quantified this relationship: parameter expansion from 100B to 1T correlates with a 17-fold increase in computational load (measured in FLOPs) and a 10-fold rise in energy consumption [125]. This translates to training costs surging from millions to tens of millions of dollars, exacerbating technological monopolization—only a handful of global institutions can sustain trillion-parameter model development. Concurrently, inference latency increases by 3–5 fold for 1T-parameter models compared to their 100B-parameter counterparts, impairing real-time applicability in edge devices or low-bandwidth environments [125].

Existing efficiency optimizations present distinct trade-offs: (MoE) architectures reduce computational demands by activating selective parameters but incur communication overheads in distributed training, limiting actual efficiency gains to 60% of theoretical potential [126]. These challenges necessitate paradigm shifts beyond scale-centric growth, emphasizing architectural innovation and training paradigm reform to rebalance efficiency and performance.

5 Future trends and prospects

5.1 Technology integration and innovation

The integration of large-scale models with quantum computing, reinforcement learning, and other state-of-the-art technologies signals profound transformations and immense potential for advancing artificial intelligence (AI) capabilities, though significant technical and ethical challenges must be addressed.

Quantum Computing Integration represents a paradigm shift addressing computational bottlenecks in LLMs. The quantum advantage in tensor network optimization could dramatically accelerate training processes; Aizpurua, B. et al. demonstrate that tasks previously requiring years might be completed in months [127]. However, current NISQ devices limit immediate application, making hybrid quantum-classical architectures the interim solution. Long-term, quantum systems may enable training of more than 10T parameter LLMs with significantly reduced carbon footprints [127]. These technological integrations confront substantial challenges including hardware limitations [128], algorithm optimization requirements [129], and data privacy concerns.

Reinforcement learning fusion constitutes another transformative trend that enhances LLMs' decision-making

capabilities by combining sequential optimization with semantic pattern extraction. The complementary nature of these capabilities makes such integration particularly promising for developing more autonomous and adaptive AI systems. Yan et al. show that scaling reinforcement learning with LLMs enables more sophisticated reasoning and planning capabilities [130]. This fusion achieves more efficient learning processes, enhances model generalization capability, and enables strong performance across a broader environments. Applications in robotics control, autonomous driving, and strategic game playing demonstrate the practical benefits of this integration.

Emerging Hybrid Architectures combine the strengths of multiple AI paradigms to overcome individual limitations. In cross-modal fields, neuro-symbolic integration merges neural networks' pattern recognition with symbolic AI's logical reasoning, reducing text-image semantic deviation by 20%; quantum-enhanced multimodal models leverage tensor network optimization to cut training time for 10B-parameter models by 40%, addressing computational bottlenecks in audio-visual fusion tasks (Section 3.4). Future research will focus on three directions: (1) hybrid quantum-neural architectures for cross-modal alignment, aiming to reduce training time by 60% while cutting semantic deviation to below 5%; (2) self-supervised multimodal pre-training with limited domain data ($\leq 50k$ pairs), targeting 90% of BioBERT's performance in low-resource medical tasks; (3) lightweight MoE designs for edge devices, reducing model size by 70% without sacrificing cross-modal alignment accuracy. Neuro-symbolic integration, for instance, merges the pattern recognition strengths of neural networks with the explicit reasoning capabilities of symbolic AI, addressing current limitations in logical consistency and explainability. Similarly, retrieval-augmented generation architectures enhance factual accuracy by grounding LLMs responses in verifiable external knowledge sources, mitigating hallucination issues that plague purely generative approaches.

Security and Privacy Considerations remain paramount throughout these technological integrations. Advanced data encryption technologies and privacy protection algorithms must be implemented to guarantee data security and user privacy [131, 132]. Zhao and Song advance differential privacy techniques specifically designed for LLMs [132]. These security foundations are essential for responsible deployment of integrated AI systems across sensitive domains including healthcare, finance, and personal assistance.

The successful realization of these technological integrations requires coordinated advances across multiple fronts: hardware development must keep pace with computational demands, algorithmic innovations must improve efficiency and robustness, and ethical frameworks must ensure beneficial outcomes. Through interdisciplinary collaboration and

sustained investment, these integrated technologies promise to unlock new capabilities in artificial intelligence while addressing current limitations in efficiency, reasoning, and safety.

Technology integration and innovation are leading the wave of scientific and technological development at an unprecedented speed. In this context, the combination of large-scale model, quantum computing [127], reinforcement learning and other cutting-edge technologies undoubtedly indicates profound changes and unlimited possibilities.

5.2 Ethics and social impact

The development of LLMs has raised critical ethical dilemmas and far-reaching societal impacts, with empirical evidence increasingly documenting their multifaceted effects on privacy, fairness, and labor markets.

Empirical studies highlight growing privacy vulnerabilities in LLMs systems. A 2024 investigation by the Electronic Privacy Information Center (EPIC) filed a complaint with the U.S. Federal Trade Commission (FTC) on October 29, 2024, calling for an investigation into OpenAI. The complaint stated that OpenAI had failed to meet the public policy standards for the responsible use of artificial intelligence and was alleged to have issues related to security, privacy protection, and business practices. Differential privacy implementations, while promising, show mixed real-world efficacy: a comparative study by Li et al. revealed that standard noise-injection techniques reduce model accuracy by 11%–15% in healthcare diagnostic tasks, creating a trade-off between privacy preservation and functional utility [133].

Systematic biases in LLMs outputs are well-documented across domains. In criminal justice risk assessment, ProPublica's 2023 analysis found that Decoder-Only models showed 23% higher false positive rates for minority defendants compared to those of majority groups, mirroring historical biases in training data [134]. Financial applications exhibit similar disparities: Ji et al. (2023) demonstrated that credit scoring models trained on historical loan data rejected applications from low-income neighborhoods at a 28% higher rate, despite controlling for creditworthiness metrics, due to spurious correlations between geographic features and default predictions [135]. These biases persist despite fairness-aware training: a 2024 benchmark study showed that even state-of-the-art debiasing techniques reduce but do not eliminate disparity, with residual gaps of 5%–8% in sensitive attribute prediction [78].

The impact of LLMs on employment follows a dual pattern of displacement and creation. A McKinsey Global Institute report estimates that 15–30% of low-skill administrative tasks (e.g., data entry, basic customer service) could be automated by 2030, disproportionately affecting workers

with less than a high school education [136]. Conversely, demand for AI-specialized roles has surged: LinkedIn's 2024 Jobs on the Rise report shows a 75% annual growth in postings for LLMs ethics consultants and multimodal data engineers, with median salaries 40% higher than traditional software roles [137].

Concerns about algorithmic echo chambers are validated by user behavior studies. A 2023 experiment by Stanford's Computational Social Science Lab found that users interacting with LLMs-powered recommendation systems encountered 47% fewer diverse viewpoints compared to traditional search engines, as models prioritize engagement-maximizing content over informational diversity [138]. This effect is amplified in educational settings: teacher surveys indicate that 61% of students rely exclusively on LLMs-generated summaries for coursework, reducing exposure to primary sources and critical analysis [139].

5.3 Policies and regulations

Governments and international bodies are advancing targeted frameworks to tackle AI's ethical, legal, and societal challenges, with emerging standards rooted in empirical risk assessments that reflect regional priorities and global coordination needs.

The EU AI Act (2024) adopts a risk-based governance model, categorizing high-risk AI applications—such as healthcare diagnostics, biometric identification, and critical infrastructure management—under strict regulatory requirements. It mandates pre-deployment conformity assessments, post-market monitoring, and transparent disclosure of algorithmic decision-making processes to mitigate biases and ensure accountability, while explicitly banning unacceptable-risk systems like mass surveillance tools [140]. This framework aligns with broader efforts to balance innovation and safeguards, though it faces challenges in harmonizing enforcement across EU member states, a gap partially addressed by guidance on cross-border compliance [141].

In the U.S., the regulatory landscape remains fragmented, with federal and state-level actions complementing each other. The NIST AI Risk Management Framework (2023) provides voluntary guidelines for identifying and mitigating AI risks—including bias in hiring or healthcare algorithms—while states like Colorado and California have introduced targeted laws: Colorado's AI Act (2024) focuses on algorithmic discrimination in high-risk sectors, and California's SB-942 (2026) mandates transparency for generative AI [140]. This decentralized approach fosters adaptability but creates inconsistencies in oversight, which frameworks for interagency coordination seek to resolve [141].

China's centralized governance model centers on aligning AI development with social stability and national security.

Its Interim Measures for Generative AI Services (2023) require real-time monitoring of AI-generated content to ensure alignment with societal goals, and high-risk systems (e.g., facial recognition) demand government registration and algorithmic audits [141]. This model enables efficient implementation of safeguards but raises questions about transparency, which recent updates to data governance rules aim to address [140]. International standardization efforts, led by bodies like ISO, play a key role in bridging regional differences. Standards focusing on AI bias assessment, risk management, and ethical principles provide a foundation for global interoperability, though their voluntary nature limits enforceability in jurisdictions with binding regulations [140]. To address this, proposals for region-specific annexes to standards have emerged, ensuring alignment with diverse regulatory priorities while preserving core ethical principles [141]. These ethical, social, and policy dimensions highlight the need for interdisciplinary approaches: combining technical innovations (e.g., bias mitigation tools) with institutional safeguards (e.g., regulatory sandboxes) and stakeholder engagement to guide responsible AI development. Strengthening the adaptability of global standards and enhancing cross-regional dialogue will be critical to fostering trust and consistency in AI governance [140].

6 Conclusion

The core innovation of this survey resides in developing an integrated analytical framework of “scenario constraints–technical selection–cost balance”, rather than merely synthesizing existing research. Unlike prior surveys that either list technologies in isolation or describe applications superficially, this work quantifies the performance boundaries and resource consumption of different technical routes—guiding researchers to prioritize high-impact directions (e.g., optimizing RLHF's alignment–efficiency trade-off) and assisting practitioners in selecting scenario-specific solutions (e.g., quantization for edge devices, Encoder-Decoder for translation tasks). This framework addresses the fundamental deficiency of prior reviews: failing to link technical characteristics to practical decision-making. This integrated framework synthesizes the current landscape of LLM research while offering a structured methodology to evaluate emerging technologies, ensuring that future technical advancements are assessed against both performance improvements and practical constraints. It also delves into post-GPT-4 key directions (multimodal fusion, RAG, AI Agent) to highlight their complementarity with traditional LLMs—e.g., multimodal models addressing modal bias via Encoder-Only's bidirectional attention, AI Agent optimizing tool-use via RLHF. This synthesis of technologies, trends,

and constraints' makes it both a technical summary and a forward-looking guide for LLMs' sustainable development.

As a cornerstone of artificial intelligence, LLMs have made significant strides in recent years, demonstrating robust capabilities in traditional fields such as natural language processing, computer vision, and speech recognition. Their applications have expanded into emerging domains including healthcare, fintech, and education, creating diverse practical scenarios that drive societal progress.

This survey systematically reviews the technical evolution of LLMs, from foundational Transformer architectures and attention mechanism variants to advanced training paradigms like RLHF. It clarifies the strengths and limitations of different architectural types (Encoder-Only, Encoder-Decoder, Decoder-Only) and summarizes effective strategies for addressing core challenges, such as data scarcity (via FSL and transfer learning), computational inefficiency (through model compression and lightweight techniques), and poor interpretability (using attention visualization and model distillation).

Despite these advancements, LLMs still confront critical challenges: high data annotation costs, factual hallucination in knowledge-intensive tasks, algorithmic bias perpetuated by imbalanced data, and a scalability trade-off between model scale and computational efficiency. Ethical and social impacts, including privacy risks, employment transformations, and information cocoon effects, further require multi-stakeholder collaboration to resolve.

Looking forward, three interrelated research directions hold particular promise for advancing LLMs toward more reliable, capable, and responsible systems. First, trustworthy LLMs should be prioritized to mitigate hallucination and bias—this involves integrating RAG with causal reasoning to ground generation in verifiable knowledge (reducing academic hallucination rates below 5%), and developing dynamic bias monitoring systems (e.g., real-time calibration of credit scoring models) to maintain fairness in high-stakes applications. Second, causal reasoning integration will address the limitations of statistical pattern matching: future work can explore neuro-symbolic architectures that combine neural networks' pattern recognition with symbolic logic's causal inference, enhancing LLMs' ability to explain decisions (e.g., linking medical diagnosis conclusions to specific image features) and generalize to unseen scenarios. Third, safety-critical applications in medicine demand targeted optimization—this includes refining domain-adaptive pre-training (DAP) for low-resolution medical imaging, integrating dual safeguards (domain knowledge bases + physician review) to cut clinical hallucination risks by over 60%, and developing lightweight multimodal models that balance diagnostic accuracy with edge-device deployability. Beyond these, the integration of LLMs with frontier

technologies like quantum computing will continue to break computational bottlenecks, while strengthened policy regulations and cross-disciplinary cooperation will ensure ethical governance. With sustained innovation and collaborative governance, LLMs are poised to play an increasingly vital role across more fields, contributing significantly to human societal progress.

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Declarations

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